

1 Equations in ρ coordinates

$$0 = \frac{1}{V'} \frac{\partial}{\partial \rho} \left[V' \langle |\nabla \rho|^2 \rangle \frac{\partial}{\partial \rho} \Phi \right] + \frac{1}{\epsilon_0} (-en_e + Z_i en_i + Z_z en_z + Z_b en_b + Z_b e g n_b^{\text{rp}}) \quad (1.1)$$

$$\begin{aligned} \frac{1}{c^2} \frac{\partial}{\partial t} \dot{\psi}_t &= V' \left\langle \frac{|\nabla \rho|^2}{R^2} \right\rangle \frac{\partial}{\partial \rho} \left[\frac{1}{V' \langle R^{-2} \rangle} \frac{\partial}{\partial \rho} \psi_t \right] \\ &+ \frac{\mu_0}{\langle B^2 \rangle} \left[-en_e \left(\langle Bu_{e\parallel} \rangle - I \left\langle \frac{u_{e\zeta}}{R} \right\rangle \right) + Z_i en_i \left(\langle Bu_{i\parallel} \rangle - I \left\langle \frac{u_{i\zeta}}{R} \right\rangle \right) \right. \\ &\left. + Z_z en_z \left(\langle Bu_{z\parallel} \rangle - I \left\langle \frac{u_{z\zeta}}{R} \right\rangle \right) + Z_b en_b \left(\langle Bu_{b\parallel} \rangle - I \left\langle \frac{u_{b\zeta}}{R} \right\rangle \right) \right] \end{aligned} \quad (1.2)$$

$$\begin{aligned} \frac{1}{c^2} \frac{\partial}{\partial t} \dot{\psi} &= \frac{1}{V' \langle R^{-2} \rangle} \frac{\partial}{\partial \rho} \left[V' \left\langle \frac{|\nabla \rho|^2}{R^2} \right\rangle \frac{\partial}{\partial \rho} \psi \right] \\ &- \frac{\mu_0}{\langle R^{-2} \rangle} \left(-en_e \left\langle \frac{u_{e\zeta}}{R} \right\rangle + Z_i en_i \left\langle \frac{u_{i\zeta}}{R} \right\rangle + Z_z en_z \left\langle \frac{u_{z\zeta}}{R} \right\rangle + Z_b en_b \left\langle \frac{u_{b\zeta}}{R} \right\rangle \right) \end{aligned} \quad (1.3)$$

$$\frac{\partial}{\partial t} \psi = \dot{\psi} \quad (1.4)$$

$$\frac{\partial}{\partial t} \psi_t = \dot{\psi}_t \quad (1.5)$$

$$\begin{aligned} \frac{1}{V'} \frac{\partial}{\partial t} (V' n_e) &= -\frac{1}{V'} \frac{\partial}{\partial \rho} [V' n_e (u_e^p - u_g^p)] + \nu_{\text{ionize}} \frac{n_e}{n_{01} + n_{02} + n_{03}} (n_{01} + n_{02} + n_{03}) \\ &+ Z_z n_e \bar{n}_z f_5 S_5 - Z_z n_e n_z A_6 - \nu_L (n_e - n_{\text{esup}}) + (1 - f_{\text{CX}}) \frac{P_b}{E_b} \end{aligned} \quad (1.6)$$

$$\begin{aligned} 0 &= -(\langle B^2 \rangle \langle R^2 \rangle - I^2) \frac{\partial}{\partial \rho} n_e T_e + en_e (\langle B^2 \rangle \langle R^2 \rangle - I^2) \frac{\partial}{\partial \rho} \Phi \\ &- e \frac{\partial \psi}{\partial \rho} I n_e \langle Bu_{e\parallel} \rangle + e \frac{\partial \psi}{\partial \rho} \langle B^2 \rangle n_e \langle Ru_{e\zeta} \rangle \end{aligned} \quad (1.7)$$

$$\begin{aligned} m_e n_e \frac{\partial}{\partial t} \langle Bu_{e\parallel} \rangle &= -\hat{\mu}_1^e \langle Bu_{e\parallel} \rangle - \hat{\mu}_2^e \langle B \hat{q}_{e\parallel} \rangle + \hat{\mu}_1^e B V_{1e} + \hat{\mu}_2^e B V_{2e} \\ &+ \ell_{11}^{\text{ee}} \langle Bu_{e\parallel} \rangle + \ell_{11}^{\text{ei}} \langle Bu_{i\parallel} \rangle + \ell_{11}^{\text{ez}} \langle Bu_{z\parallel} \rangle + \ell_{11}^{\text{eb}} \langle Bu_{b\parallel} \rangle \\ &- \ell_{12}^{\text{ee}} \langle B \hat{q}_{e\parallel} \rangle - \ell_{12}^{\text{ei}} \langle B \hat{q}_{i\parallel} \rangle - \ell_{12}^{\text{ez}} \langle B \hat{q}_{z\parallel} \rangle \\ &- en_e \langle BE_{\parallel} \rangle - \nu_L m_e n_e \langle Bu_{e\parallel} \rangle - \nu_{0e} m_e n_e \langle Bu_{e\parallel} \rangle \\ &- \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho| \rangle v_{e\zeta} m_e n_e \langle Bu_{e\parallel} \rangle + \langle |\nabla \rho| \rangle v_{e\zeta} m_e n_e \left(\frac{\partial}{\partial \rho} \frac{\langle B^2 \rangle}{I} \right) \frac{I}{\langle B^2 \rangle} \langle Bu_{e\parallel} \rangle \end{aligned}$$

$$\begin{aligned}
& + \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle \chi_{e\zeta} m_e n_e \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial \rho} \frac{I}{\langle B^2 \rangle} \langle Bu_{e\parallel} \rangle \\
& - \langle |\nabla \rho|^2 \rangle \chi_{e\zeta} m_e n_e \left(\frac{\partial}{\partial \rho} \frac{\langle B^2 \rangle}{I} \right) \frac{\partial}{\partial \rho} \frac{I}{\langle B^2 \rangle} \langle Bu_{e\parallel} \rangle - \frac{\langle B^2 \rangle}{I} \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle \Pi_e^{\text{res}} \rangle \\
& - \frac{1}{V'} \frac{\partial}{\partial \rho} V' \left[\langle |\nabla \rho|^2 \rangle v_{e\zeta} m_e n_e \left(\langle R^2 \rangle - \frac{I^2}{\langle B^2 \rangle} \right) \frac{1}{I} BV_{1e} \right. \\
& \left. - \langle |\nabla \rho|^2 \rangle \chi_{e\zeta} m_e n_e \frac{\partial}{\partial \rho} \left(\langle R^2 \rangle - \frac{I^2}{\langle B^2 \rangle} \right) \frac{1}{I} BV_{1e} \right] \tag{1.8}
\end{aligned}$$

$$\begin{aligned}
\frac{1}{V'} \frac{\partial}{\partial t} (V' m_e n_e \langle Ru_{e\zeta} \rangle) & = - \frac{1}{V'} \frac{\partial}{\partial \rho} V' \left[\langle |\nabla \rho|^2 \rangle v_{e\zeta} m_e n_e \langle Ru_{e\zeta} \rangle + (u_e^\rho - u_g^\rho) m_e n_e \langle Ru_{e\zeta} \rangle \right. \\
& \left. - \langle |\nabla \rho|^2 \rangle \chi_{e\zeta} m_e n_e \frac{\partial}{\partial \rho} \langle Ru_{e\zeta} \rangle + \langle \Pi_e^{\text{res}} \rangle \right] \\
& + \ell_{11}^{\text{ee}} \langle Ru_{e\zeta} \rangle + \ell_{11}^{\text{ei}} \langle Ru_{i\zeta} \rangle + \ell_{11}^{\text{ez}} \langle Ru_{z\zeta} \rangle + \ell_{11}^{\text{eb}} \langle Ru_{b\zeta} \rangle \\
& - \ell_{12}^{\text{ee}} \langle R\hat{q}_{e\zeta} \rangle - \ell_{12}^{\text{ei}} \langle R\hat{q}_{i\zeta} \rangle - \ell_{12}^{\text{ez}} \langle R\hat{q}_{z\zeta} \rangle \\
& - en_e \langle RE_\zeta \rangle - e \frac{\partial \psi}{\partial \rho} n_e u_e^\rho \\
& - \frac{e \langle |\nabla \rho|^2 \rangle D_e}{T_e} \frac{\partial \psi}{\partial \rho} \left[(1 - C_T) T_e \frac{\partial}{\partial \rho} n_e + C_T \frac{\partial}{\partial \rho} n_e T_e + n_e T_e \frac{\partial \psi}{\partial \rho} C_P \right] \\
& - \nu_L m_e n_e \langle Ru_{e\zeta} \rangle - \nu_{0e} m_e n_e \langle Ru_{e\zeta} \rangle \tag{1.9}
\end{aligned}$$

$$\begin{aligned}
\frac{3}{2} \frac{1}{V'^{5/3}} \frac{\partial}{\partial t} (V'^{5/3} n_e T_e) & = - \frac{1}{V'} \frac{\partial}{\partial \rho} V' \left[- \chi_e \langle |\nabla \rho|^2 \rangle n_e \frac{\partial}{\partial \rho} T_e + V_e^{\text{hp}} \langle |\nabla \rho|^2 \rangle n_e T_e + \frac{5}{2} n_e T_e (u_e^\rho - u_g^\rho) \right] \\
& - u_g^\rho \frac{\partial}{\partial \rho} n_e T_e - u_i^\rho \frac{\partial}{\partial \rho} n_i T_i - u_z^\rho \frac{\partial}{\partial \rho} n_z T_z - \hat{u}_{i\theta} \langle \mathbf{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_i \rangle - \hat{u}_{z\theta} \langle \mathbf{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_z \rangle \\
& - \frac{3}{2} n_e \frac{T_e - T_i}{\tau_{ei}} - \frac{3}{2} n_e \frac{T_e - T_z}{\tau_{ez}} \\
& - \dot{\psi}_t \langle B^\theta \rangle (-en_e \hat{u}_{e\theta} + Z_i en_i \hat{u}_{i\theta} + Z_z en_z \hat{u}_{z\theta}) \\
& + \dot{\psi} \left(-en_e \left\langle \frac{u_{e\zeta}}{R} \right\rangle + Z_i en_i \left\langle \frac{u_{i\zeta}}{R} \right\rangle + Z_z en_z \left\langle \frac{u_{z\zeta}}{R} \right\rangle \right) \\
& - \nu_L T_e (n_e - n_{\text{esup}}) - \frac{3}{2} \nu_{LT_e} n_e (T_e - T_{\text{esup}}) \\
& - E_H \nu_{\text{ionize}} \frac{n_e}{n_{01} + n_{02} + n_{03}} (n_{01} + n_{02} + n_{03}) \\
& + f_{ei} P_b + P_{ae} + P_{RFe} - P_{Br} \tag{1.10}
\end{aligned}$$

$$\frac{5}{2} m_e n_e \frac{\partial}{\partial t} \langle B\hat{q}_{e\parallel} \rangle = -\hat{\mu}_2^e \langle Bu_{e\parallel} \rangle - \hat{\mu}_3^e \langle B\hat{q}_{e\parallel} \rangle + \hat{\mu}_2^e BV_{1e} + \hat{\mu}_3^e BV_{2e}$$

$$\begin{aligned}
& -\ell_{21}^{\text{ee}}\langle Bu_{\text{e}\parallel}\rangle - \ell_{21}^{\text{ei}}\langle Bu_{\text{i}\parallel}\rangle - \ell_{21}^{\text{ez}}\langle Bu_{\text{z}\parallel}\rangle - \ell_{21}^{\text{eb}}\langle Bu_{\text{b}\parallel}\rangle \\
& + \ell_{22}^{\text{ee}}\langle B\hat{q}_{\text{e}\parallel}\rangle + \ell_{22}^{\text{ei}}\langle B\hat{q}_{\text{i}\parallel}\rangle + \ell_{22}^{\text{ez}}\langle B\hat{q}_{\text{z}\parallel}\rangle \\
& - \nu_{\text{L}}m_{\text{e}}n_{\text{e}}\langle B\hat{q}_{\text{e}\parallel}\rangle - \nu_{0\text{e}}m_{\text{e}}n_{\text{e}}\langle B\hat{q}_{\text{e}\parallel}\rangle
\end{aligned} \tag{1.11}$$

$$\begin{aligned}
\frac{1}{V'}\frac{\partial}{\partial t}(V'n_{\text{i}}) &= -\frac{1}{V'}\frac{\partial}{\partial\rho}[V'n_{\text{i}}(u_{\text{i}}^{\rho} - u_{\text{g}}^{\rho})] + \nu_{\text{ionize}}\frac{n_{\text{e}}}{n_{01} + n_{02} + n_{03}}(n_{01} + n_{02} + n_{03}) \\
& - \nu_{\text{L}}(n_{\text{i}} - n_{\text{isup}}) + \nu_{\text{b}}n_{\text{b}} + \nu_{\text{b}}gn_{\text{b}}^{\text{rp}} - f_{\text{CX}}\frac{P_{\text{b}}}{E_{\text{b}}} + \nu_{\text{bL}}n_{\text{b}} + S_{\text{LC}} - \nu_{\text{OL}}n_{\text{i}}
\end{aligned} \tag{1.12}$$

$$\begin{aligned}
0 &= -(\langle B^2\rangle\langle R^2\rangle - I^2)\frac{\partial}{\partial\rho}n_{\text{i}}T_{\text{i}} - Z_{\text{i}}en_{\text{i}}(\langle B^2\rangle\langle R^2\rangle - I^2)\frac{\partial}{\partial\rho}\Phi \\
& + Z_{\text{i}}en_{\text{i}}\frac{\partial\psi}{\partial\rho}I\langle Bu_{\text{i}\parallel}\rangle - Z_{\text{i}}en_{\text{i}}\frac{\partial\psi}{\partial\rho}\langle B^2\rangle\langle Ru_{\text{i}\zeta}\rangle
\end{aligned} \tag{1.13}$$

$$\begin{aligned}
m_{\text{i}}n_{\text{i}}\frac{\partial}{\partial t}\langle Bu_{\text{i}\parallel}\rangle &= -\hat{\mu}_1^{\text{i}}\langle Bu_{\text{i}\parallel}\rangle - \hat{\mu}_2^{\text{i}}\langle B\hat{q}_{\text{i}\parallel}\rangle + \hat{\mu}_1^{\text{i}}BV_{1\text{i}} + \hat{\mu}_2^{\text{i}}BV_{2\text{i}} \\
& + \ell_{11}^{\text{ie}}\langle Bu_{\text{e}\parallel}\rangle + \ell_{11}^{\text{ii}}\langle Bu_{\text{i}\parallel}\rangle + \ell_{11}^{\text{iz}}\langle Bu_{\text{z}\parallel}\rangle + \ell_{11}^{\text{ib}}\langle Bu_{\text{b}\parallel}\rangle \\
& - \ell_{12}^{\text{ie}}\langle B\hat{q}_{\text{e}\parallel}\rangle - \ell_{12}^{\text{ii}}\langle B\hat{q}_{\text{i}\parallel}\rangle - \ell_{12}^{\text{iz}}\langle B\hat{q}_{\text{z}\parallel}\rangle \\
& + Z_{\text{i}}en_{\text{i}}\langle BE_{\parallel}\rangle - \nu_{\text{L}}m_{\text{i}}n_{\text{i}}\langle Bu_{\text{i}\parallel}\rangle + \nu_{\text{b}}m_{\text{b}}n_{\text{b}}\langle Bu_{\text{b}\parallel}\rangle - \nu_{0\text{i}}m_{\text{i}}n_{\text{i}}\langle Bu_{\text{i}\parallel}\rangle \\
& - \nu_{\text{CX}}m_{\text{i}}n_{\text{i}}\langle Bu_{\text{i}\parallel}\rangle - \nu_{\text{OL}}m_{\text{i}}n_{\text{i}}\langle Bu_{\text{i}\parallel}\rangle \\
& - \frac{1}{V'}\frac{\partial}{\partial\rho}V'\langle|\nabla\rho|\rangle v_{\text{i}\zeta}m_{\text{i}}n_{\text{i}}\langle Bu_{\text{i}\parallel}\rangle + \langle|\nabla\rho|\rangle v_{\text{i}\zeta}m_{\text{i}}n_{\text{i}}\left(\frac{\partial}{\partial\rho}\frac{\langle B^2\rangle}{I}\right)\frac{I}{\langle B^2\rangle}\langle Bu_{\text{i}\parallel}\rangle \\
& + \frac{1}{V'}\frac{\partial}{\partial\rho}V'\langle|\nabla\rho|^2\rangle\chi_{\text{i}\zeta}m_{\text{i}}n_{\text{i}}\frac{\langle B^2\rangle}{I}\frac{\partial}{\partial\rho}\frac{I}{\langle B^2\rangle}\langle Bu_{\text{i}\parallel}\rangle \\
& - \langle|\nabla\rho|^2\rangle\chi_{\text{i}\zeta}m_{\text{i}}n_{\text{i}}\left(\frac{\partial}{\partial\rho}\frac{\langle B^2\rangle}{I}\right)\frac{\partial}{\partial\rho}\frac{I}{\langle B^2\rangle}\langle Bu_{\text{i}\parallel}\rangle - \frac{\langle B^2\rangle}{I}\frac{1}{V'}\frac{\partial}{\partial\rho}V'\langle\Pi_{\text{i}}^{\text{res}}\rangle \\
& - \frac{1}{V'}\frac{\partial}{\partial\rho}V'\left[\langle|\nabla\rho|\rangle v_{\text{i}\zeta}m_{\text{i}}n_{\text{i}}\left(\langle R^2\rangle - \frac{I^2}{\langle B^2\rangle}\right)\frac{1}{I}BV_{1\text{i}}\right. \\
& \left. - \langle|\nabla\rho|^2\rangle\chi_{\text{i}\zeta}m_{\text{i}}n_{\text{i}}\frac{\partial}{\partial\rho}\left(\langle R^2\rangle - \frac{I^2}{\langle B^2\rangle}\right)\frac{1}{I}BV_{1\text{i}}\right]
\end{aligned} \tag{1.14}$$

$$\begin{aligned}
\frac{1}{V'}\frac{\partial}{\partial t}(V'm_{\text{i}}n_{\text{i}}\langle Ru_{\text{i}\zeta}\rangle) &= -\frac{1}{V'}\frac{\partial}{\partial\rho}V'\left[\langle|\nabla\rho|\rangle v_{\text{i}\zeta}m_{\text{i}}n_{\text{i}}\langle Ru_{\text{i}\zeta}\rangle + (u_{\text{i}}^{\rho} - u_{\text{g}}^{\rho})m_{\text{i}}n_{\text{i}}\langle Ru_{\text{i}\zeta}\rangle\right. \\
& \left. - \langle|\nabla\rho|^2\rangle\chi_{\text{i}\zeta}m_{\text{i}}n_{\text{i}}\frac{\partial}{\partial\rho}\langle Ru_{\text{i}\zeta}\rangle + \langle\Pi_{\text{i}}^{\text{res}}\rangle\right] \\
& + \ell_{11}^{\text{ie}}\langle Ru_{\text{e}\zeta}\rangle + \ell_{11}^{\text{ii}}\langle Ru_{\text{i}\zeta}\rangle + \ell_{11}^{\text{iz}}\langle Ru_{\text{z}\zeta}\rangle + \ell_{11}^{\text{ib}}\langle Ru_{\text{b}\zeta}\rangle
\end{aligned}$$

$$\begin{aligned}
& -\ell_{12}^{\text{ie}}\langle R\hat{q}_{\text{e}\zeta}\rangle - \ell_{12}^{\text{ii}}\langle R\hat{q}_{\text{i}\zeta}\rangle - \ell_{12}^{\text{iz}}\langle R\hat{q}_{\text{z}\zeta}\rangle \\
& + Z_{\text{i}}en_{\text{i}}\langle RE_{\zeta}\rangle + Z_{\text{i}}e\frac{\partial\psi}{\partial\rho}n_{\text{i}}u_{\text{i}}^{\rho} \\
& + Z_{\text{i}}\frac{e\langle|\nabla\rho|^2\rangle D_{\text{e}}}{T_{\text{e}}}\frac{\partial\psi}{\partial\rho}\frac{n_{\text{i}}+n_{\text{b}}}{n_{\text{e}}}\left[(1-C_{\text{T}})T_{\text{e}}\frac{\partial}{\partial\rho}n_{\text{e}}+C_{\text{T}}\frac{\partial}{\partial\rho}n_{\text{e}}T_{\text{e}}+n_{\text{e}}T_{\text{e}}\frac{\partial\psi}{\partial\rho}C_{\text{p}}\right] \\
& -\nu_{\text{L}}m_{\text{i}}n_{\text{i}}\langle Ru_{\text{i}\zeta}\rangle + \nu_{\text{b}}m_{\text{b}}n_{\text{b}}\langle Ru_{\text{b}\zeta}\rangle - \nu_{0\text{i}}m_{\text{i}}n_{\text{i}}\langle Ru_{\text{i}\zeta}\rangle - \nu_{\text{CX}}m_{\text{i}}n_{\text{i}}\langle Ru_{\text{i}\zeta}\rangle \\
& -\nu_{\text{OL}}m_{\text{i}}n_{\text{i}}\langle Ru_{\text{i}\zeta}\rangle
\end{aligned} \tag{1.15}$$

$$\begin{aligned}
\frac{3}{2}\frac{1}{V'^{5/3}}\frac{\partial}{\partial t}(V'^{5/3}n_{\text{i}}T_{\text{i}}) &= -\frac{1}{V'}\frac{\partial}{\partial\rho}V'\left[-\chi_{\text{i}}\langle|\nabla\rho|^2\rangle n_{\text{i}}\frac{\partial}{\partial\rho}T_{\text{i}}+V_{\text{i}}^{\text{hp}}\langle|\nabla\rho|\rangle n_{\text{i}}T_{\text{i}}+\frac{5}{2}n_{\text{i}}T_{\text{i}}(u_{\text{i}}^{\rho}-u_{\text{g}}^{\rho})\right] \\
&+ (u_{\text{i}}^{\rho}-u_{\text{g}}^{\rho})\frac{\partial}{\partial\rho}n_{\text{i}}T_{\text{i}}+\hat{u}_{\text{i}\theta}\langle\mathbf{B}\cdot\nabla\cdot\overleftrightarrow{\pi}_{\text{i}}\rangle-\frac{3}{2}n_{\text{i}}\frac{T_{\text{i}}-T_{\text{e}}}{\tau_{\text{ie}}}-\frac{3}{2}n_{\text{i}}\frac{T_{\text{i}}-T_{\text{z}}}{\tau_{\text{iz}}} \\
&- \nu_{\text{L}}T_{\text{i}}(n_{\text{i}}-n_{\text{isup}})-\frac{3}{2}\nu_{\text{LT}_{\text{i}}}n_{\text{i}}(T_{\text{i}}-T_{\text{isup}}) \\
&+ \frac{3}{2}\nu_{\text{ionize}}\frac{n_{\text{e}}}{n_{01}+n_{02}+n_{03}}(n_{01}T_{01}+n_{02}T_{02}+n_{03}T_{03}) \\
&- \frac{3}{2}\nu_{\text{CX}}n_{\text{i}}(T_{\text{i}}-T_{01})+(1-f_{\text{ei}})P_{\text{b}}+m_{\text{b}}\frac{P_{\text{b}}}{E_{\text{b}}}\langle Bu_{\text{b}\parallel}\rangle\frac{\langle Bu_{\text{i}\parallel}\rangle}{\langle B^2\rangle}+P_{\text{ai}}+P_{\text{RFi}}
\end{aligned} \tag{1.16}$$

$$\begin{aligned}
\frac{5}{2}m_{\text{i}}n_{\text{i}}\frac{\partial}{\partial t}\langle B\hat{q}_{\text{i}\parallel}\rangle &= -\hat{\mu}_2^{\text{i}}\langle Bu_{\text{i}\parallel}\rangle - \hat{\mu}_3^{\text{i}}\langle B\hat{q}_{\text{i}\parallel}\rangle + \hat{\mu}_2^{\text{i}}BV_{1\text{i}} + \hat{\mu}_3^{\text{i}}BV_{2\text{i}} \\
&- \ell_{21}^{\text{ie}}\langle Bu_{\text{e}\parallel}\rangle - \ell_{21}^{\text{ii}}\langle Bu_{\text{i}\parallel}\rangle - \ell_{21}^{\text{iz}}\langle Bu_{\text{z}\parallel}\rangle - \ell_{21}^{\text{ib}}\langle Bu_{\text{b}\parallel}\rangle \\
&+ \ell_{22}^{\text{ie}}\langle B\hat{q}_{\text{e}\parallel}\rangle + \ell_{22}^{\text{ii}}\langle B\hat{q}_{\text{i}\parallel}\rangle + \ell_{22}^{\text{iz}}\langle B\hat{q}_{\text{z}\parallel}\rangle \\
&- \nu_{\text{L}}m_{\text{i}}n_{\text{i}}\langle B\hat{q}_{\text{i}\parallel}\rangle - \nu_{0\text{i}}m_{\text{i}}n_{\text{i}}\langle B\hat{q}_{\text{i}\parallel}\rangle
\end{aligned} \tag{1.17}$$

$$\begin{aligned}
\frac{1}{V'}\frac{\partial}{\partial t}(V'n_{\text{z}}) &= -\frac{1}{V'}\frac{\partial}{\partial\rho}[V'n_{\text{z}}(u_{\text{z}}^{\rho}-u_{\text{g}}^{\rho})]+n_{\text{e}}\bar{n}_{\text{z}}f_5S_5-n_{\text{e}}n_{\text{z}}A_6 \\
&- \nu_{\text{L}}(n_{\text{z}}-n_{\text{zsup}})+S_{\text{LC}}-\nu_{\text{OL}}n_{\text{z}}
\end{aligned} \tag{1.18}$$

$$\begin{aligned}
0 &= -(\langle B^2\rangle\langle R^2\rangle-I^2)\frac{\partial}{\partial\rho}n_{\text{z}}T_{\text{z}}-Z_{\text{z}}en_{\text{z}}(\langle B^2\rangle\langle R^2\rangle-I^2)\frac{\partial}{\partial\rho}\Phi \\
&+ Z_{\text{z}}en_{\text{z}}\frac{\partial\psi}{\partial\rho}I\langle Bu_{\text{z}\parallel}\rangle-Z_{\text{z}}en_{\text{z}}\frac{\partial\psi}{\partial\rho}\langle B^2\rangle\langle Ru_{\text{z}\zeta}\rangle
\end{aligned} \tag{1.19}$$

$$\begin{aligned}
m_{\text{z}}n_{\text{z}}\frac{\partial}{\partial t}\langle Bu_{\text{z}\parallel}\rangle &= -\hat{\mu}_1^{\text{z}}\langle Bu_{\text{z}\parallel}\rangle - \hat{\mu}_2^{\text{z}}\langle B\hat{q}_{\text{z}\parallel}\rangle + \hat{\mu}_1^{\text{z}}BV_{1\text{z}} + \hat{\mu}_2^{\text{z}}BV_{2\text{z}} \\
&+ \ell_{11}^{\text{ze}}\langle Bu_{\text{e}\parallel}\rangle + \ell_{11}^{\text{zi}}\langle Bu_{\text{i}\parallel}\rangle + \ell_{11}^{\text{zz}}\langle Bu_{\text{z}\parallel}\rangle + \ell_{11}^{\text{zb}}\langle Bu_{\text{b}\parallel}\rangle \\
&- \ell_{12}^{\text{ze}}\langle B\hat{q}_{\text{e}\parallel}\rangle - \ell_{12}^{\text{zi}}\langle B\hat{q}_{\text{i}\parallel}\rangle - \ell_{12}^{\text{zz}}\langle B\hat{q}_{\text{z}\parallel}\rangle
\end{aligned}$$

$$\begin{aligned}
& + Z_z e n_z \langle BE_{\parallel} \rangle - \nu_L m_z n_z \langle Bu_{z\parallel} \rangle - \nu_{0z} m_z n_z \langle Bu_{z\parallel} \rangle - \nu_{OL} m_z n_z \langle Bu_{z\parallel} \rangle \\
& - \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho| \rangle v_{z\zeta} m_z n_z \langle Bu_{z\parallel} \rangle + \langle |\nabla \rho| \rangle v_{z\zeta} m_z n_z \left(\frac{\partial \langle B^2 \rangle}{\partial \rho} \frac{I}{I} \right) \frac{I}{\langle B^2 \rangle} \langle Bu_{z\parallel} \rangle \\
& + \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle \chi_{z\zeta} m_z n_z \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial \rho} \frac{I}{\langle B^2 \rangle} \langle Bu_{z\parallel} \rangle \\
& - \langle |\nabla \rho|^2 \rangle \chi_{z\zeta} m_z n_z \left(\frac{\partial \langle B^2 \rangle}{\partial \rho} \frac{I}{I} \right) \frac{\partial}{\partial \rho} \frac{I}{\langle B^2 \rangle} \langle Bu_{z\parallel} \rangle - \frac{\langle B^2 \rangle}{I} \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle \Pi_z^{\text{res}} \rangle \\
& - \frac{1}{V'} \frac{\partial}{\partial \rho} V' \left[\langle |\nabla \rho| \rangle v_{z\zeta} m_z n_z \left(\langle R^2 \rangle - \frac{I^2}{\langle B^2 \rangle} \right) \frac{1}{I} B V_{1z} \right. \\
& \left. - \langle |\nabla \rho|^2 \rangle \chi_{z\zeta} m_z n_z \frac{\partial}{\partial \rho} \left(\langle R^2 \rangle - \frac{I^2}{\langle B^2 \rangle} \right) \frac{1}{I} B V_{1z} \right] \tag{1.20}
\end{aligned}$$

$$\begin{aligned}
\frac{1}{V'} \frac{\partial}{\partial t} (V' m_z n_z \langle Ru_{z\zeta} \rangle) & = - \frac{1}{V'} \frac{\partial}{\partial \rho} V' \left[\langle |\nabla \rho| \rangle v_{z\zeta} m_z n_z \langle Ru_{z\zeta} \rangle + (u_z^\rho - u_g^\rho) m_z n_z \langle Ru_{z\zeta} \rangle \right. \\
& \left. - \langle |\nabla \rho|^2 \rangle \chi_{z\zeta} m_z n_z \frac{\partial}{\partial \rho} \langle Ru_{z\zeta} \rangle + \langle \Pi_z^{\text{res}} \rangle \right] \\
& + \ell_{11}^{\text{ze}} \langle Ru_{e\zeta} \rangle + \ell_{11}^{\text{zi}} \langle Ru_{i\zeta} \rangle + \ell_{11}^{\text{zz}} \langle Ru_{z\zeta} \rangle + \ell_{11}^{\text{zb}} \langle Ru_{b\zeta} \rangle \\
& - \ell_{12}^{\text{ze}} \langle R\hat{q}_{e\zeta} \rangle - \ell_{12}^{\text{zi}} \langle R\hat{q}_{i\zeta} \rangle - \ell_{12}^{\text{zz}} \langle R\hat{q}_{z\zeta} \rangle \\
& + Z_z e n_z \langle RE_\zeta \rangle + Z_z e \frac{\partial \psi}{\partial \rho} n_z u_z^\rho \\
& + Z_z \frac{e \langle |\nabla \rho|^2 \rangle D_e}{T_e} \frac{\partial \psi}{\partial \rho} \frac{n_z}{n_e} \left[(1 - C_T) T_e \frac{\partial}{\partial \rho} n_e + C_T \frac{\partial}{\partial \rho} n_e T_e + n_e T_e \frac{\partial \psi}{\partial \rho} C_p \right] \\
& - \nu_L m_z n_z \langle Ru_{z\zeta} \rangle - \nu_{0z} m_z n_z \langle Ru_{z\zeta} \rangle - \nu_{OL} m_z n_z \langle Ru_{z\zeta} \rangle \tag{1.21}
\end{aligned}$$

$$\begin{aligned}
\frac{3}{2} \frac{1}{V'^{5/3}} \frac{\partial}{\partial t} (V'^{5/3} n_z T_z) & = - \frac{1}{V'} \frac{\partial}{\partial \rho} V' \left[- \chi_z \langle |\nabla \rho|^2 \rangle n_z \frac{\partial}{\partial \rho} T_z + V_z^{\text{hp}} \langle |\nabla \rho| \rangle n_z T_z + \frac{5}{2} n_z T_z (u_z^\rho - u_g^\rho) \right] \\
& + (u_z^\rho - u_g^\rho) \frac{\partial}{\partial \rho} n_z T_z + \hat{u}_{z\theta} \langle \mathbf{B} \cdot \nabla \cdot \overset{\leftrightarrow}{\pi}_z \rangle - \frac{3}{2} n_z \frac{T_z - T_e}{\tau_{ze}} - \frac{3}{2} n_z \frac{T_z - T_i}{\tau_{zi}} \\
& - \nu_L T_z (n_z - n_{z\text{sup}}) - \frac{3}{2} \nu_{LT_z} n_z (T_z - T_{z\text{sup}}) \\
& + P_{\alpha z} + P_{\text{RF}z} \tag{1.22}
\end{aligned}$$

$$\begin{aligned}
\frac{5}{2} m_z n_z \frac{\partial}{\partial t} \langle B\hat{q}_{z\parallel} \rangle & = - \hat{\mu}_2^z \langle Bu_{z\parallel} \rangle - \hat{\mu}_3^z \langle B\hat{q}_{z\parallel} \rangle + \hat{\mu}_2^z B V_{1z} + \hat{\mu}_3^z B V_{2z} \\
& - \ell_{21}^{\text{ze}} \langle Bu_{e\parallel} \rangle - \ell_{21}^{\text{zi}} \langle Bu_{i\parallel} \rangle - \ell_{21}^{\text{zz}} \langle Bu_{z\parallel} \rangle - \ell_{21}^{\text{zb}} \langle Bu_{b\parallel} \rangle \\
& + \ell_{22}^{\text{ze}} \langle B\hat{q}_{e\parallel} \rangle + \ell_{22}^{\text{zi}} \langle B\hat{q}_{i\parallel} \rangle + \ell_{22}^{\text{zz}} \langle B\hat{q}_{z\parallel} \rangle
\end{aligned}$$

$$- \nu_L m_z n_z \langle B \hat{q}_{z\parallel} \rangle - \nu_{0z} m_z n_z \langle B \hat{q}_{z\parallel} \rangle \quad (1.23)$$

$$\begin{aligned} \frac{1}{V'} \frac{\partial}{\partial t} (V' n_b) &= -\frac{1}{V'} \frac{\partial}{\partial \rho} [V' n_b (u_b^\rho - u_g^\rho)] + \frac{P_b}{E_b} - g \frac{P_{b\perp}}{E_b} - \nu_b n_b - \nu_{bL} n_b - \nu_{\text{eff}} g (\sqrt{\delta} n_b - n_b^{\text{rp}}) \\ &+ \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle f_t D_b \frac{\partial}{\partial \rho} n_b \end{aligned} \quad (1.24)$$

$$\begin{aligned} 0 &= -(\langle B^2 \rangle \langle R^2 \rangle - I^2) \frac{\partial}{\partial \rho} n_b T_b - Z_b e n_b (\langle B^2 \rangle \langle R^2 \rangle - I^2) \frac{\partial}{\partial \rho} \Phi \\ &+ Z_b e n_b \frac{\partial \psi}{\partial \rho} I \langle B u_{b\parallel} \rangle - Z_b e n_b \frac{\partial \psi}{\partial \rho} \langle B^2 \rangle \langle R u_{b\zeta} \rangle \end{aligned} \quad (1.25)$$

$$\begin{aligned} m_b \frac{\partial}{\partial t} \langle B u_{b\parallel} \rangle &= -(\hat{\mu}_1^b - \ell_{11}^{bb}) \langle B u_{b\parallel} \rangle + \sum_{a \neq b} \ell_{11}^{ba} \langle B u_{a\parallel} \rangle + \hat{\mu}_1^b B V_{1b} + Z_b e n_b \langle B E_{\parallel}^A \rangle \\ &- \nu_{bL} m_b n_b \langle B u_{b\parallel} \rangle - \nu_b m_b n_b \langle B u_{b\parallel} \rangle - \nu_{0b} m_b n_b \langle B u_{b\parallel} \rangle - \nu_{CX} m_b n_b \langle B u_{b\parallel} \rangle + \langle B S_{m\parallel b} \rangle \\ &- \nu_{\text{eff}} g \sqrt{\delta} m_b n_b \langle B u_{b\parallel} \rangle + \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle m_b \langle B u_{b\parallel} \rangle f_t D_b \frac{\partial}{\partial \rho} n_b \end{aligned} \quad (1.26)$$

$$\begin{aligned} \frac{1}{V'} \frac{\partial}{\partial t} (V' m_b n_b \langle R u_{b\zeta} \rangle) &= -\frac{1}{V'} \frac{\partial}{\partial \rho} [V' (u_b^\rho - u_g^\rho) m_b n_b \langle R u_{b\zeta} \rangle] \\ &+ \ell_{11}^{be} \langle R u_{e\zeta} \rangle + \ell_{11}^{bi} \langle R u_{i\zeta} \rangle + \ell_{11}^{bz} \langle R u_{z\zeta} \rangle + \ell_{11}^{bb} n_b \langle R u_{b\zeta} \rangle \\ &+ Z_b e n_b \dot{\psi} + Z_b e \frac{\partial \psi}{\partial \rho} n_b u_b^\rho - \nu_{bL} m_b n_b \langle R u_{b\zeta} \rangle - \nu_b m_b n_b \langle B u_{b\zeta} \rangle \\ &- \nu_{0i} m_b n_b \langle R u_{b\zeta} \rangle - \nu_{CX} m_b n_b \langle R u_{b\zeta} \rangle + \frac{I}{\langle B^2 \rangle} \langle B S_{m\parallel b} \rangle \end{aligned} \quad (1.27)$$

$$\begin{aligned} \frac{1}{V'} \frac{\partial}{\partial t} (V' n_{01}) &= \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle D_{01} \frac{\partial}{\partial \rho} n_{01} - \nu_{\text{ionize}} \frac{n_e}{n_{01} + n_{02} + n_{03}} n_{01} \\ &- \nu_{CX} \frac{n_i}{n_{01} + n_{02}} n_{01} + \gamma_0 \nu_L (n_i - n_{\text{isup}}) + \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle \mathbf{\Gamma}_i^{\text{puff}} \cdot \nabla \rho \rangle \Big|_b \end{aligned} \quad (1.28)$$

$$\frac{1}{V'} \frac{\partial}{\partial t} (V' n_{02}) = \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle D_{02} \frac{\partial}{\partial \rho} n_{02} - \nu_{\text{ionize}} \frac{n_e}{n_{01} + n_{02} + n_{03}} n_{02} + \nu_{CX} \frac{n_i}{n_{01} + n_{02}} n_{01} \quad (1.29)$$

$$\frac{1}{V'} \frac{\partial}{\partial t} (V' n_{03}) = \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle D_{03} \frac{\partial}{\partial \rho} n_{03} - \nu_{\text{ionize}} \frac{n_e}{n_{01} + n_{02} + n_{03}} n_{03} + f_{CX} \frac{P_b}{E_b} \quad (1.30)$$

$$\begin{aligned} \frac{1}{V'} \frac{\partial}{\partial t} (V' \bar{n}_z) &= \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle D_{0z} \frac{\partial}{\partial \rho} \bar{n}_z - n_e \bar{n}_z f_5 S_5 + n_e n_z A_6 \\ &+ \gamma_{0z} \nu_L (n_z - n_{z\text{sup}}) + \gamma_{i,z} \nu_L (n_i - n_{\text{isup}}) + \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle \mathbf{\Gamma}_z^{\text{puff}} \cdot \nabla \rho \rangle \Big|_b \end{aligned} \quad (1.31)$$

$$\frac{1}{V'} \frac{\partial}{\partial t} (V' n_b^{\text{rp}}) = \frac{P_{b\perp}}{E_b} + \nu_{\text{eff}}(\sqrt{\delta} n_b - n_b^{\text{rp}}) - \nu_b n_b^{\text{rp}} - \frac{1}{V'} \frac{\partial}{\partial \rho} V' \left(\langle |\nabla \rho| \rangle u_b^{\text{rp}} n_b^{\text{rp}} - \langle |\nabla \rho|^2 \rangle D_b^{\text{rp}} \frac{\partial}{\partial \rho} n_b^{\text{rp}} \right) \quad (1.32)$$

2 Additional important relationship

$$\bar{n}_z = \sum_{i=0}^{\max(Z)-1} n_i \quad (2.1)$$

$$B_p = \frac{|\nabla \rho|}{R} \frac{\partial \psi}{\partial \rho}, \quad \langle B_p^2 \rangle = \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \left(\frac{\partial \psi}{\partial V} \right)^2 \quad (2.2)$$

$$B_t = \frac{I}{R} = \frac{1}{R} \frac{4\pi^2}{\langle R^{-2} \rangle} \frac{\partial \psi_t}{\partial V}, \quad \langle B_t^2 \rangle = \frac{16\pi^4}{\langle R^{-2} \rangle} \left(\frac{\partial \psi_t}{\partial V} \right)^2 \quad (2.3)$$

$$\langle B^2 \rangle = \langle B_t^2 \rangle + \langle B_p^2 \rangle = \frac{16\pi^4}{\langle R^{-2} \rangle} \left(\frac{\partial \psi_t}{\partial V} \right)^2 + \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \left(\frac{\partial \psi}{\partial V} \right)^2 \quad (2.4)$$

$$\langle B^\theta \rangle = \frac{4\pi^2}{V'} \frac{\partial \psi}{\partial \rho} = 4\pi^2 \frac{\partial \psi}{\partial V} = \frac{I \langle R^{-2} \rangle}{q} = 2\pi \left(\oint \frac{dl_p}{B_p} \right)^{-1} \quad (2.5)$$

$$\hat{q}^2 \equiv \frac{I^2}{2\langle B_p^2 \rangle} \left(\left\langle \frac{1}{R^2} \right\rangle - \frac{1}{\langle R^2 \rangle} \right) \quad (2.6)$$

$$\langle B^2 \rangle \langle R^2 \rangle - I^2 = \langle R^2 \rangle \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \left(\frac{\partial \psi}{\partial V} \right)^2 (1 + 2\hat{q}^2) \quad (2.7)$$

$$\langle B \hat{q}_{s\parallel} \rangle \equiv \frac{2\langle B q_{s\parallel} \rangle}{5p_s} \quad (2.8)$$

$$\hat{u}_{s\theta} = \frac{\langle B u_{s\parallel} \rangle - B V_{1s}}{\langle B^2 \rangle} \quad (2.9)$$

$$\begin{aligned} \left\langle \frac{u_{s\zeta}}{R} \right\rangle &= \frac{1}{1 + 2\hat{q}^2} \frac{1}{\langle R^2 \rangle} \langle R u_{s\zeta} \rangle + \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{I} \langle B u_{s\parallel} \rangle \\ &= \frac{\langle R u_{s\zeta} \rangle}{\langle R^2 \rangle} + \frac{I}{\langle B^2 \rangle} \left(\left\langle \frac{1}{R^2} \right\rangle - \frac{1}{\langle R^2 \rangle} \right) (\langle B u_{s\parallel} \rangle - B V_{1s}) \\ &= \frac{\langle R u_{s\zeta} \rangle}{\langle R^2 \rangle} + I \left(\left\langle \frac{1}{R^2} \right\rangle - \frac{1}{\langle R^2 \rangle} \right) \hat{u}_{s\theta} \end{aligned} \quad (2.10)$$

$$\langle R \hat{q}_{s\zeta} \rangle = \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{1}{I} B V_{2s} + \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{s\parallel} \rangle \quad (2.11)$$

$$B V_{1s} = -I \left(\frac{1}{e_s n_s} \frac{\partial n_s T_s}{\partial \psi} + \frac{\partial \Phi}{\partial \psi} \right), \quad B V_{2s} = -\frac{I}{e_s} \frac{\partial T_s}{\partial \psi} \quad (2.12)$$

$$u^\rho \equiv \langle \mathbf{u} \cdot \nabla \rho \rangle, \quad u^V = \langle \mathbf{u} \cdot \nabla V \rangle \quad (2.13)$$

$$u_g^\rho = -\frac{4\pi^2}{IV'\langle R^{-2} \rangle} \dot{\psi}_t, \quad u_g^V = -\frac{4\pi^2}{I\langle R^{-2} \rangle} \dot{\psi}_t = -\frac{V'}{2\rho\psi_{ta}} \dot{\psi}_t \quad (2.14)$$

$$\frac{1}{V'} \frac{\partial V'}{\partial t} = \frac{\partial}{\partial V} \langle \mathbf{u}_g \cdot \nabla V \rangle = \frac{\partial u_g^V}{\partial V} \quad (2.15)$$

$$\langle E_r \rangle = -\langle |\nabla \rho|^2 \rangle \frac{\partial r}{\partial \rho} \frac{\partial}{\partial \rho} \Phi \quad (2.16)$$

$$\langle BE_{\parallel} \rangle = I\langle R^{-2} \rangle \left(-\frac{\dot{\psi}_t}{q} + \dot{\psi} \right) \quad (2.17)$$

$$RE_\zeta = \dot{\psi} = \frac{V_{\text{loop}}}{2\pi} \quad (2.18)$$

$$\langle \mathbf{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_i \rangle = \hat{\mu}_1^i (\langle Bu_{i\parallel} \rangle - BV_{1i}) + \hat{\mu}_2^i \left(\frac{2\langle Bq_{i\parallel} \rangle}{5p_i} - BV_{2i} \right) \quad (2.19)$$

$$\hat{\ell}_{ij}^{ab} \equiv \frac{\ell_{ij}^{ab}}{m_a n_a}, \quad \hat{\ell}_{ij}^{bb} \equiv \frac{\ell_{ij}^{bb}}{m_e n_e} \quad (2.20)$$

$$\Gamma^{\text{puff}} \simeq \frac{\langle \mathbf{\Gamma}^{\text{puff}} \cdot \nabla \rho \rangle}{\langle |\nabla \rho| \rangle} \quad (2.21)$$

3 Equations in the coordinates suitable for implementation

$$0 = \frac{\partial}{\partial V} \left[\langle |\nabla V|^2 \rangle \frac{\partial}{\partial V} \Phi \right] + \frac{1}{\epsilon_0} (-e n_e + Z_i e n_i + Z_z e n_z + Z_b e n_b + Z_b e g n_b^{\text{rp}}) \quad (3.1)$$

$$\begin{aligned} \epsilon_0 \frac{\partial}{\partial t} \dot{\psi}_t &= \frac{\partial}{\partial V} \left[\left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{1}{\langle R^{-2} \rangle} \frac{\partial}{\partial V} \frac{\psi_t}{\mu_0} \right] - \frac{1}{\langle R^{-2} \rangle} \frac{\partial}{\partial V} \left[\left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \right] \frac{\partial}{\partial V} \frac{\psi_t}{\mu_0} \\ &+ \frac{1}{\langle B\theta \rangle} \left[-e \left(n_e \langle Bu_{e\parallel} \rangle - I n_e \left\langle \frac{u_{e\zeta}}{R} \right\rangle \right) + Z_i e \left(n_i \langle Bu_{i\parallel} \rangle - I n_i \left\langle \frac{u_{i\zeta}}{R} \right\rangle \right) \right. \\ &\left. + Z_z e \left(n_z \langle Bu_{z\parallel} \rangle - I n_z \left\langle \frac{u_{z\zeta}}{R} \right\rangle \right) + Z_b e \left(n_b \langle Bu_{b\parallel} \rangle - I n_b \left\langle \frac{u_{b\zeta}}{R} \right\rangle \right) \right] \end{aligned} \quad (3.2)$$

$$\begin{aligned} \frac{1}{c^2} \left\langle \frac{1}{R^2} \right\rangle \frac{\partial}{\partial t} \dot{\psi} &= \frac{\partial}{\partial V} \left[\left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{\partial}{\partial V} \psi \right] \\ &- \mu_0 \left(-e n_e \left\langle \frac{u_{e\zeta}}{R} \right\rangle + Z_i e n_i \left\langle \frac{u_{i\zeta}}{R} \right\rangle + Z_z e n_z \left\langle \frac{u_{z\zeta}}{R} \right\rangle + Z_b e n_b \left\langle \frac{u_{b\zeta}}{R} \right\rangle \right) \end{aligned} \quad (3.3)$$

$$\frac{\partial}{\partial t} \psi = \dot{\psi} \quad (3.4)$$

$$\frac{\partial}{\partial t} \frac{\psi_t}{\mu_0} = \frac{\dot{\psi}_t}{\mu_0} \quad (3.5)$$

$$\begin{aligned} \frac{\partial}{\partial t} n_e = & -\frac{\partial}{\partial V} n_e u_e^V + u_g^V \frac{\partial}{\partial V} n_e + \nu_{\text{ionize}} \frac{n_e}{n_{01} + n_{02} + n_{03}} (n_{01} + n_{02} + n_{03}) \\ & + Z_z n_e \bar{n}_z f_5 S_5 - Z_z n_e n_z A_6 - \nu_L (n_e - n_{\text{esup}}) + (1 - f_{\text{CX}}) \frac{P_b}{E_b} \end{aligned} \quad (3.6)$$

$$0 = -\frac{e}{m_e} n_e \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{1}{I} B V_{1e} - \frac{e}{m_e} I n_e \langle B u_{e\parallel} \rangle + \frac{e}{m_e} \langle B^2 \rangle n_e \langle R u_{e\zeta} \rangle \quad (3.7)$$

$$\begin{aligned} n_e \frac{\partial}{\partial t} \langle B u_{e\parallel} \rangle = & -\frac{1}{m_e} \hat{\mu}_1^e \langle B u_{e\parallel} \rangle - \frac{1}{m_e} \hat{\mu}_2^e \langle B \hat{q}_{e\parallel} \rangle + \frac{1}{m_e} \hat{\mu}_1^e B V_{1e} \\ & + \frac{1}{em_e} \hat{\mu}_2^e I \frac{\partial V}{\partial \psi} \frac{1}{n_e} \frac{\partial}{\partial V} n_e T_e - \frac{1}{em_e} \hat{\mu}_2^e I \frac{\partial V}{\partial \psi} \frac{T_e}{n_e} \frac{\partial}{\partial V} n_e \\ & + \hat{\ell}_{11}^{ee} n_e \langle B u_{e\parallel} \rangle + \hat{\ell}_{11}^{ei} n_e \langle B u_{i\parallel} \rangle + \hat{\ell}_{11}^{ez} n_e \langle B u_{z\parallel} \rangle + \hat{\ell}_{11}^{eb} \frac{n_e}{n_b} n_b \langle B u_{b\parallel} \rangle \\ & - \hat{\ell}_{12}^{ee} n_e \langle B \hat{q}_{e\parallel} \rangle - \hat{\ell}_{12}^{ei} n_e \langle B \hat{q}_{i\parallel} \rangle - \hat{\ell}_{12}^{ez} n_e \langle B \hat{q}_{z\parallel} \rangle \\ & - e \frac{n_e}{m_e} I \langle R^{-2} \rangle \left(-\frac{\dot{\psi}_t}{q} + \dot{\psi} \right) - \nu_L n_e \langle B u_{e\parallel} \rangle - \nu_{0e} n_e \langle B u_{e\parallel} \rangle \\ & - \frac{\partial}{\partial V} \langle |\nabla \rho| \rangle v_{e\zeta} n_e \langle B u_{e\parallel} \rangle + \langle |\nabla V| \rangle v_{e\zeta} n_e \left(\frac{\partial}{\partial V} \frac{\langle B^2 \rangle}{I} \right) \frac{I}{\langle B^2 \rangle} \langle B u_{e\parallel} \rangle \\ & + \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{e\zeta} n_e \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial V} \frac{I}{\langle B^2 \rangle} \langle B u_{e\parallel} \rangle \\ & - \langle |\nabla V|^2 \rangle \chi_{e\zeta} n_e \left(\frac{\partial}{\partial V} \frac{\langle B^2 \rangle}{I} \right) \frac{\partial}{\partial V} \frac{I}{\langle B^2 \rangle} \langle B u_{e\parallel} \rangle - \frac{1}{m_e} \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial V} \langle \Pi_e^{\text{res}} \rangle \\ & - \frac{\partial}{\partial V} \left[\langle |\nabla V| \rangle v_{e\zeta} n_e \frac{\langle R^2 \rangle}{\langle B^2 \rangle} \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{\partial \psi}{\partial V} (1 + 2\hat{q}^2) \frac{1}{I} \frac{\partial \psi}{\partial V} B V_{1e} \right. \\ & \left. - \langle |\nabla V|^2 \rangle \chi_{e\zeta} n_e \frac{\partial}{\partial V} \frac{\langle R^2 \rangle}{\langle B^2 \rangle} \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{\partial \psi}{\partial V} (1 + 2\hat{q}^2) \frac{1}{I} \frac{\partial \psi}{\partial V} B V_{1e} \right] \end{aligned} \quad (3.8)$$

$$\begin{aligned} \frac{\partial}{\partial t} n_e \langle R u_{e\zeta} \rangle = & -(u_e^V - u_g^V) \frac{\partial}{\partial V} n_e \langle R u_{e\zeta} \rangle - \frac{\partial u_e^V}{\partial V} n_e \langle R u_{e\zeta} \rangle - \frac{\partial}{\partial V} [\langle |\nabla V| \rangle v_{e\zeta} n_e \langle R u_{e\zeta} \rangle] \\ & + \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{e\zeta} \frac{\partial}{\partial V} n_e \langle R u_{e\zeta} \rangle - \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{e\zeta} \langle R u_{e\zeta} \rangle \frac{\partial}{\partial V} n_e - \frac{1}{m_e} \frac{\partial}{\partial V} \langle \Pi_e^{\text{res}} \rangle \\ & + \hat{\ell}_{11}^{ee} n_e \langle R u_{e\zeta} \rangle + \hat{\ell}_{11}^{ei} \frac{n_e}{n_i} n_i \langle R u_{i\zeta} \rangle + \hat{\ell}_{11}^{ez} \frac{n_e}{n_z} n_z \langle R u_{z\zeta} \rangle + \hat{\ell}_{11}^{eb} \frac{n_e}{n_b} n_b \langle R u_{b\zeta} \rangle \end{aligned}$$

$$\begin{aligned}
& -\hat{\ell}_{12}^{\text{ee}} n_{\text{e}} \frac{1}{e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\text{e}} - \hat{\ell}_{12}^{\text{ee}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\text{e}\parallel} \rangle \\
& + \hat{\ell}_{12}^{\text{ei}} n_{\text{e}} \frac{1}{Z_{\text{i}} e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\text{i}} - \hat{\ell}_{12}^{\text{ei}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\text{i}\parallel} \rangle \\
& + \hat{\ell}_{12}^{\text{ez}} n_{\text{e}} \frac{1}{Z_{\text{z}} e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\text{z}} - \hat{\ell}_{12}^{\text{ez}} n_{\text{e}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\text{z}\parallel} \rangle \\
& - e \frac{n_{\text{e}}}{m_{\text{e}}} \dot{\psi} - e \frac{1}{m_{\text{e}}} \frac{\partial \psi}{\partial V} n_{\text{e}} u_{\text{e}}^V \\
& - \frac{e^2 \langle |\nabla V|^2 \rangle D_{\text{e}}}{m_{\text{e}} T_{\text{e}}} \frac{\partial \psi}{\partial V} \left[n_{\text{e}} \frac{\partial \psi}{\partial V} \frac{1}{I} B V_{1\text{e}} + n_{\text{e}} \frac{\partial}{\partial V} \Phi + \frac{C_{\text{T}} - 1}{e} \left(\frac{\partial}{\partial V} n_{\text{e}} T_{\text{e}} - T_{\text{e}} \frac{\partial}{\partial V} n_{\text{e}} \right) + \frac{T_{\text{e}}}{e} \frac{\partial \psi}{\partial V} C_{\text{p}} n_{\text{e}} \right] \\
& - \nu_{\text{L}} n_{\text{e}} \langle R u_{\text{e}\zeta} \rangle - \nu_{0\text{e}} n_{\text{e}} \langle R u_{\text{e}\zeta} \rangle
\end{aligned} \tag{3.9}$$

$$\begin{aligned}
\frac{3}{2} \frac{\partial}{\partial t} n_{\text{e}} T_{\text{e}} & = -\frac{5}{2} (u_{\text{e}}^V - u_{\text{g}}^V) \frac{\partial}{\partial V} n_{\text{e}} T_{\text{e}} - \frac{5}{2} \frac{\partial u_{\text{e}}^V}{\partial V} n_{\text{e}} T_{\text{e}} - \frac{\partial}{\partial V} [V_{\text{e}}^{\text{hp}} \langle |\nabla V| \rangle n_{\text{e}} T_{\text{e}}] \\
& + \frac{\partial}{\partial V} \chi_{\text{e}} \langle |\nabla V|^2 \rangle \frac{\partial}{\partial V} n_{\text{e}} T_{\text{e}} - \frac{\partial}{\partial V} \chi_{\text{e}} \langle |\nabla V|^2 \rangle T_{\text{e}} \frac{\partial}{\partial V} n_{\text{e}} \\
& - u_{\text{g}}^V \frac{\partial}{\partial V} n_{\text{e}} T_{\text{e}} - u_{\text{i}}^V \frac{\partial}{\partial V} n_{\text{i}} T_{\text{i}} - u_{\text{z}}^V \frac{\partial}{\partial V} n_{\text{z}} T_{\text{z}} - \hat{u}_{\text{i}\theta} \langle \mathbf{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_{\text{i}} \rangle - \hat{u}_{\text{z}\theta} \langle \mathbf{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_{\text{z}} \rangle \\
& - \frac{3}{2} \nu_{\text{Tei}} \left(n_{\text{e}} T_{\text{e}} - \frac{n_{\text{e}}}{n_{\text{i}}} n_{\text{i}} T_{\text{i}} \right) - \frac{3}{2} \nu_{\text{Tez}} \left(n_{\text{e}} T_{\text{e}} - \frac{n_{\text{e}}}{n_{\text{z}}} n_{\text{z}} T_{\text{z}} \right) \\
& - 4\pi^2 \frac{\partial \psi}{\partial V} \dot{\psi}_t (-e n_{\text{e}} \hat{u}_{\text{e}\theta} + Z_{\text{i}} e n_{\text{i}} \hat{u}_{\text{i}\theta} + Z_{\text{z}} e n_{\text{z}} \hat{u}_{\text{z}\theta}) \\
& + \dot{\psi} \left(-e n_{\text{e}} \left\langle \frac{u_{\text{e}\zeta}}{R} \right\rangle + Z_{\text{i}} e n_{\text{i}} \left\langle \frac{u_{\text{i}\zeta}}{R} \right\rangle + Z_{\text{z}} e n_{\text{z}} \left\langle \frac{u_{\text{z}\zeta}}{R} \right\rangle \right) \\
& - \nu_{\text{L}} (n_{\text{e}} T_{\text{e}} - n_{\text{esup}} T_{\text{e}}) - \frac{3}{2} \nu_{\text{LT}_{\text{e}}} (n_{\text{e}} T_{\text{e}} - n_{\text{e}} T_{\text{esup}}) - E_{\text{H}} \nu_{\text{ionize}} \frac{n_{\text{e}}}{n_{01} + n_{02} + n_{03}} (n_{01} + n_{02} + n_{03}) \\
& + f_{\text{ei}} P_{\text{b}} + P_{\alpha\text{e}} + P_{\text{RFe}} - P_{\text{Br}}
\end{aligned} \tag{3.10}$$

$$\begin{aligned}
\frac{5}{2} n_{\text{e}} \frac{\partial}{\partial t} \langle B \hat{q}_{\text{e}\parallel} \rangle & = -\frac{1}{m_{\text{e}}} \hat{\mu}_2^{\text{e}} \langle B u_{\text{e}\parallel} \rangle - \frac{1}{m_{\text{e}}} \hat{\mu}_3^{\text{e}} \langle B \hat{q}_{\text{e}\parallel} \rangle + \frac{1}{m_{\text{e}}} \hat{\mu}_2^{\text{e}} B V_{1\text{e}} \\
& + \frac{1}{e m_{\text{e}}} \hat{\mu}_3^{\text{e}} I \frac{\partial V}{\partial \psi} \frac{1}{n_{\text{e}}} \frac{\partial}{\partial V} n_{\text{e}} T_{\text{e}} - \frac{1}{e m_{\text{e}}} \hat{\mu}_3^{\text{e}} I \frac{\partial V}{\partial \psi} \frac{T_{\text{e}}}{n_{\text{e}}} \frac{\partial}{\partial V} n_{\text{e}} \\
& - \hat{\ell}_{21}^{\text{ee}} n_{\text{e}} \langle B u_{\text{e}\parallel} \rangle - \hat{\ell}_{21}^{\text{ei}} n_{\text{e}} \langle B u_{\text{i}\parallel} \rangle - \hat{\ell}_{21}^{\text{ez}} n_{\text{e}} \langle B u_{\text{z}\parallel} \rangle - \hat{\ell}_{21}^{\text{eb}} \frac{n_{\text{e}}}{n_{\text{b}}} n_{\text{b}} \langle B u_{\text{b}\parallel} \rangle \\
& + \hat{\ell}_{22}^{\text{ee}} n_{\text{e}} \langle B \hat{q}_{\text{e}\parallel} \rangle + \hat{\ell}_{22}^{\text{ei}} n_{\text{e}} \langle B \hat{q}_{\text{i}\parallel} \rangle + \hat{\ell}_{22}^{\text{ez}} n_{\text{e}} \langle B \hat{q}_{\text{z}\parallel} \rangle \\
& - \nu_{\text{L}} n_{\text{e}} \langle B \hat{q}_{\text{e}\parallel} \rangle - \nu_{0\text{e}} n_{\text{e}} \langle B \hat{q}_{\text{e}\parallel} \rangle
\end{aligned} \tag{3.11}$$

$$0 = -n_{\text{e}} \left\langle \frac{u_{\text{e}\zeta}}{R} \right\rangle + \frac{1}{\langle R^2 \rangle} n_{\text{e}} \langle R u_{\text{e}\zeta} \rangle - \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{\langle R^2 \rangle} n_{\text{e}} \langle R u_{\text{e}\zeta} \rangle + \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{I} n_{\text{e}} \langle B u_{\text{e}\parallel} \rangle \tag{3.12}$$

$$0 = -n_e B V_{1e} + \frac{1}{e} \frac{\partial V}{\partial \psi} I \frac{\partial}{\partial V} n_e T_e - n_e \frac{\partial V}{\partial \psi} I \frac{\partial}{\partial V} \Phi \quad (3.13)$$

$$\begin{aligned} \frac{\partial}{\partial t} n_i &= -\frac{\partial}{\partial V} n_i u_i^V + u_g^V \frac{\partial}{\partial V} n_i + \nu_{\text{ionize}} \frac{n_e}{n_{01} + n_{02} + n_{03}} (n_{01} + n_{02} + n_{03}) - \nu_L (n_i - n_{\text{isup}}) \\ &+ \nu_b n_b + \nu_b g n_b^{\text{rp}} - f_{\text{CX}} \frac{P_b}{E_b} + \nu_{bL} n_b + S_{\text{LC}} - \nu_{\text{OL}} n_i \end{aligned} \quad (3.14)$$

$$0 = Z_i \frac{e}{m_i} n_i \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{1}{I} B V_{1i} + Z_i \frac{e}{m_i} I n_i \langle B u_{i\parallel} \rangle - Z_i \frac{e}{m_i} \langle B^2 \rangle n_i \langle R u_{i\zeta} \rangle \quad (3.15)$$

$$\begin{aligned} n_i \frac{\partial}{\partial t} \langle B u_{i\parallel} \rangle &= -\frac{1}{m_i} \hat{\mu}_1^i \langle B u_{i\parallel} \rangle - \frac{1}{m_i} \hat{\mu}_2^i \langle B \hat{q}_{i\parallel} \rangle + \frac{1}{m_i} \hat{\mu}_1^i B V_{1i} \\ &- \frac{1}{Z_i e m_i} \hat{\mu}_2^i I \frac{\partial V}{\partial \psi} \frac{1}{n_i} \frac{\partial}{\partial V} n_i T_i + \frac{1}{Z_i e m_i} \hat{\mu}_2^i I \frac{\partial V}{\partial \psi} \frac{T_i}{n_i} \frac{\partial}{\partial V} n_i \\ &+ \hat{\ell}_{11}^{\text{ie}} n_i \langle B u_{e\parallel} \rangle + \hat{\ell}_{11}^{\text{ii}} n_i \langle B u_{i\parallel} \rangle + \hat{\ell}_{11}^{\text{iz}} n_i \langle B u_{z\parallel} \rangle + \hat{\ell}_{11}^{\text{ib}} \frac{n_i}{n_b} n_b \langle B u_{b\parallel} \rangle \\ &- \hat{\ell}_{12}^{\text{ie}} n_i \langle B \hat{q}_{e\parallel} \rangle - \hat{\ell}_{12}^{\text{ii}} n_i \langle B \hat{q}_{i\parallel} \rangle - \hat{\ell}_{12}^{\text{iz}} n_i \langle B \hat{q}_{z\parallel} \rangle \\ &+ Z_i e \frac{n_i}{m_i} I \langle R^{-2} \rangle \left(-\frac{\dot{\psi}_t}{q} + \dot{\psi} \right) \\ &- \nu_L n_i \langle B u_{i\parallel} \rangle + \nu_b \frac{m_b}{m_i} n_b \langle B u_{b\parallel} \rangle - \nu_{0i} n_i \langle B u_{i\parallel} \rangle - \nu_{\text{CX}} n_i \langle B u_{i\parallel} \rangle - \nu_{\text{LC}} n_i \langle B u_{i\parallel} \rangle - \nu_{\text{OL}} n_i \langle B u_{i\parallel} \rangle \\ &- \frac{\partial}{\partial V} \langle |\nabla \rho| \rangle v_{i\zeta} n_i \langle B u_{i\parallel} \rangle + \langle |\nabla V| \rangle v_{i\zeta} n_i \left(\frac{\partial}{\partial V} \frac{\langle B^2 \rangle}{I} \right) \frac{I}{\langle B^2 \rangle} \langle B u_{i\parallel} \rangle \\ &+ \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{i\zeta} n_i \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial V} \frac{I}{\langle B^2 \rangle} \langle B u_{i\parallel} \rangle \\ &- \langle |\nabla V|^2 \rangle \chi_{i\zeta} n_i \left(\frac{\partial}{\partial V} \frac{\langle B^2 \rangle}{I} \right) \frac{\partial}{\partial V} \frac{I}{\langle B^2 \rangle} \langle B u_{i\parallel} \rangle - \frac{1}{m_i} \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial V} \langle \Pi_i^{\text{res}} \rangle \\ &- \frac{\partial}{\partial V} \left[\langle |\nabla V| \rangle v_{i\zeta} n_i \frac{\langle R^2 \rangle}{\langle B^2 \rangle} \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{\partial \psi}{\partial V} (1 + 2\hat{q}^2) \frac{1}{I} \frac{\partial \psi}{\partial V} B V_{1i} \right. \\ &\left. - \langle |\nabla V|^2 \rangle \chi_{i\zeta} n_i \frac{\partial}{\partial V} \frac{\langle R^2 \rangle}{\langle B^2 \rangle} \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{\partial \psi}{\partial V} (1 + 2\hat{q}^2) \frac{1}{I} \frac{\partial \psi}{\partial V} B V_{1i} \right] \end{aligned} \quad (3.16)$$

$$\begin{aligned} \frac{\partial}{\partial t} n_i \langle R u_{i\zeta} \rangle &= -(u_i^V - u_g^V) \frac{\partial}{\partial V} n_i \langle R u_{i\zeta} \rangle - \frac{\partial u_i^V}{\partial V} n_i \langle R u_{i\zeta} \rangle - \frac{\partial}{\partial V} \langle |\nabla V| \rangle v_{i\zeta} n_i \langle R u_{i\zeta} \rangle \\ &+ \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{i\zeta} \frac{\partial}{\partial V} n_i \langle R u_{i\zeta} \rangle - \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{i\zeta} \langle R u_{i\zeta} \rangle \frac{\partial}{\partial V} n_i - \frac{1}{m_i} \frac{\partial}{\partial V} \langle \Pi_i^{\text{res}} \rangle \\ &+ \hat{\ell}_{11}^{\text{ie}} \frac{n_i}{n_e} n_e \langle R u_{e\zeta} \rangle + \hat{\ell}_{11}^{\text{ii}} n_i \langle R u_{i\zeta} \rangle + \hat{\ell}_{11}^{\text{iz}} \frac{n_i}{n_z} n_z \langle R u_{z\zeta} \rangle + \hat{\ell}_{11}^{\text{ib}} \frac{n_i}{n_b} n_b \langle R u_{b\zeta} \rangle \end{aligned}$$

$$\begin{aligned}
& -\tilde{\ell}_{12}^{\text{ie}} n_{\text{i}} \frac{1}{e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\text{e}} - \tilde{\ell}_{12}^{\text{ie}} n_{\text{i}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\text{e}\parallel} \rangle \\
& + \tilde{\ell}_{12}^{\text{ii}} n_{\text{i}} \frac{1}{Z_{\text{i}} e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\text{i}} - \tilde{\ell}_{12}^{\text{ii}} n_{\text{i}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\text{i}\parallel} \rangle \\
& + \tilde{\ell}_{12}^{\text{iz}} n_{\text{i}} \frac{1}{Z_{\text{z}} e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\text{z}} - \tilde{\ell}_{12}^{\text{iz}} n_{\text{i}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\text{z}\parallel} \rangle \\
& + Z_{\text{i}} \frac{e}{m_{\text{i}}} n_{\text{i}} \dot{\psi} + Z_{\text{i}} \frac{e}{m_{\text{i}}} \frac{\partial \psi}{\partial V} n_{\text{i}} u_{\text{i}}^V \\
& + Z_{\text{i}} \frac{e^2 \langle |\nabla V|^2 \rangle D_{\text{e}}}{m_{\text{i}} T_{\text{e}}} \frac{\partial \psi}{\partial V} \left[(n_{\text{i}} + n_{\text{b}}) \frac{\partial \psi}{\partial V} \frac{1}{I} B V_{\text{1e}} + (n_{\text{i}} + n_{\text{b}}) \frac{\partial}{\partial V} \Phi \right. \\
& \left. + \frac{C_{\text{T}} - 1}{e} \frac{n_{\text{i}} + n_{\text{b}}}{n_{\text{e}}} \left(\frac{\partial}{\partial V} n_{\text{e}} T_{\text{e}} - T_{\text{e}} \frac{\partial}{\partial V} n_{\text{e}} \right) + \frac{T_{\text{e}}}{e} \frac{\partial \psi}{\partial V} C_{\text{p}} n_{\text{i}} + \frac{T_{\text{e}}}{e} \frac{\partial \psi}{\partial V} C_{\text{p}} n_{\text{b}} \right] \\
& - \nu_{\text{L}} n_{\text{i}} \langle R u_{\text{i}\zeta} \rangle + \nu_{\text{b}} \frac{m_{\text{b}}}{m_{\text{i}}} n_{\text{b}} \langle B u_{\text{b}\zeta} \rangle - \nu_{0\text{i}} n_{\text{i}} \langle R u_{\text{i}\zeta} \rangle - \nu_{\text{CX}} n_{\text{i}} \langle R u_{\text{i}\zeta} \rangle - \nu_{\text{OL}} n_{\text{i}} \langle R u_{\text{i}\zeta} \rangle
\end{aligned} \tag{3.17}$$

$$\begin{aligned}
\frac{3}{2} \frac{\partial}{\partial t} n_{\text{i}} T_{\text{i}} &= -\frac{5}{2} (u_{\text{i}}^V - u_{\text{g}}^V) \frac{\partial}{\partial V} n_{\text{i}} T_{\text{i}} - \frac{5}{2} \frac{\partial u_{\text{i}}^V}{\partial V} n_{\text{i}} T_{\text{i}} - \frac{\partial}{\partial V} V_{\text{i}}^{\text{hp}} \langle |\nabla V| \rangle n_{\text{i}} T_{\text{i}} \\
& + \frac{\partial}{\partial V} \chi_{\text{i}} \langle |\nabla V|^2 \rangle \frac{\partial}{\partial V} n_{\text{i}} T_{\text{i}} - \frac{\partial}{\partial V} \chi_{\text{i}} \langle |\nabla V|^2 \rangle T_{\text{i}} \frac{\partial}{\partial V} n_{\text{i}} \\
& + (u_{\text{i}}^V - u_{\text{g}}^V) \frac{\partial}{\partial V} n_{\text{i}} T_{\text{i}} + \hat{u}_{\text{i}\theta} \langle \mathbf{B} \cdot \nabla \cdot \vec{\pi}_{\text{i}} \rangle - \frac{3}{2} \nu_{\text{Tei}} \left(\frac{n_{\text{e}}}{n_{\text{i}}} n_{\text{i}} T_{\text{i}} - n_{\text{e}} T_{\text{e}} \right) - \frac{3}{2} \nu_{\text{Tiz}} \left(n_{\text{i}} T_{\text{i}} - \frac{n_{\text{i}}}{n_{\text{z}}} n_{\text{z}} T_{\text{z}} \right) \\
& - \nu_{\text{L}} (n_{\text{i}} T_{\text{i}} - n_{\text{isup}} T_{\text{i}}) - \frac{3}{2} \nu_{\text{LTi}} (n_{\text{i}} T_{\text{i}} - n_{\text{i}} T_{\text{isup}}) \\
& + \frac{3}{2} \nu_{\text{ionize}} \frac{n_{\text{e}}}{n_{01} + n_{02} + n_{03}} (n_{01} T_{01} + n_{02} T_{02} + n_{03} T_{03}) \\
& - \frac{3}{2} \nu_{\text{CX}} (n_{\text{i}} T_{\text{i}} - n_{\text{i}} T_{01}) + (1 - f_{\text{ei}}) P_{\text{b}} + m_{\text{b}} \frac{1}{n_{\text{b}}} \frac{P_{\text{b}}}{E_{\text{b}}} \frac{\langle B u_{\text{i}\parallel} \rangle}{\langle B^2 \rangle} n_{\text{b}} \langle B u_{\text{b}\parallel} \rangle + P_{\text{ai}} + P_{\text{RFi}}
\end{aligned} \tag{3.18}$$

$$\begin{aligned}
\frac{5}{2} n_{\text{i}} \frac{\partial}{\partial t} \langle B \hat{q}_{\text{i}\parallel} \rangle &= -\frac{1}{m_{\text{i}}} \hat{\mu}_2^{\text{i}} \langle B u_{\text{i}\parallel} \rangle - \frac{1}{m_{\text{i}}} \hat{\mu}_3^{\text{i}} \langle B \hat{q}_{\text{i}\parallel} \rangle + \frac{1}{m_{\text{i}}} \hat{\mu}_2^{\text{i}} B V_{\text{1i}} \\
& - \frac{1}{Z_{\text{i}} e m_{\text{i}}} \hat{\mu}_3^{\text{i}} I \frac{\partial V}{\partial \psi} \frac{1}{n_{\text{i}}} \frac{\partial}{\partial V} n_{\text{i}} T_{\text{i}} + \frac{1}{Z_{\text{i}} e m_{\text{i}}} \hat{\mu}_3^{\text{i}} I \frac{\partial V}{\partial \psi} \frac{T_{\text{i}}}{n_{\text{i}}} \frac{\partial}{\partial V} n_{\text{i}} \\
& - \tilde{\ell}_{21}^{\text{ie}} n_{\text{i}} \langle B u_{\text{e}\parallel} \rangle - \tilde{\ell}_{21}^{\text{ii}} n_{\text{i}} \langle B u_{\text{i}\parallel} \rangle - \tilde{\ell}_{21}^{\text{iz}} n_{\text{i}} \langle B u_{\text{z}\parallel} \rangle - \tilde{\ell}_{21}^{\text{ib}} \frac{n_{\text{i}}}{n_{\text{b}}} n_{\text{b}} \langle B u_{\text{b}\parallel} \rangle \\
& + \tilde{\ell}_{22}^{\text{ie}} n_{\text{i}} \langle B \hat{q}_{\text{e}\parallel} \rangle + \tilde{\ell}_{22}^{\text{ii}} n_{\text{i}} \langle B \hat{q}_{\text{i}\parallel} \rangle + \tilde{\ell}_{22}^{\text{iz}} n_{\text{i}} \langle B \hat{q}_{\text{z}\parallel} \rangle \\
& - \nu_{\text{L}} n_{\text{i}} \langle B \hat{q}_{\text{i}\parallel} \rangle - \nu_{0\text{i}} n_{\text{i}} \langle B \hat{q}_{\text{i}\parallel} \rangle
\end{aligned} \tag{3.19}$$

$$0 = -n_{\text{i}} \left\langle \frac{u_{\text{i}\zeta}}{R} \right\rangle + \frac{1}{\langle R^2 \rangle} n_{\text{i}} \langle R u_{\text{i}\zeta} \rangle - \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{\langle R^2 \rangle} n_{\text{i}} \langle R u_{\text{i}\zeta} \rangle + \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{I} n_{\text{i}} \langle B u_{\text{i}\parallel} \rangle \tag{3.20}$$

$$0 = -n_i \textcolor{red}{BV}_{1i} - \frac{1}{Z_i e} \frac{\partial V}{\partial \psi} I \frac{\partial}{\partial V} n_i \textcolor{red}{T}_i - n_i \frac{\partial V}{\partial \psi} I \frac{\partial}{\partial V} \Phi \quad (3.21)$$

$$\frac{\partial}{\partial t} n_z = -\frac{\partial}{\partial V} n_z u_z^V + u_g^V \frac{\partial}{\partial V} n_z + n_e \bar{n}_z f_5 S_5 - n_e n_z A_6 - \nu_L (n_z - n_{\text{isup}}) + S_{\text{LC}} - \nu_{\text{OL}} n_z \quad (3.22)$$

$$0 = Z_z \frac{e}{m_z} n_z \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{1}{I} \textcolor{red}{BV}_{1z} + Z_z \frac{e}{m_z} I n_z \langle \textcolor{red}{Bu}_{z\parallel} \rangle - Z_z \frac{e}{m_z} \langle B^2 \rangle n_z \langle \textcolor{red}{Ru}_{z\zeta} \rangle \quad (3.23)$$

$$\begin{aligned} n_z \frac{\partial}{\partial t} \langle \textcolor{red}{Bu}_{z\parallel} \rangle &= -\frac{1}{m_i} \hat{\mu}_1^z \langle \textcolor{red}{Bu}_{z\parallel} \rangle - \frac{1}{m_i} \hat{\mu}_2^z \langle \textcolor{red}{B}\hat{q}_{z\parallel} \rangle + \frac{1}{m_z} \hat{\mu}_1^i \textcolor{red}{BV}_{1z} \\ &\quad - \frac{1}{Z_z e m_z} \hat{\mu}_2^z I \frac{\partial V}{\partial \psi} \frac{1}{n_z} \frac{\partial}{\partial V} n_z \textcolor{red}{T}_z + \frac{1}{Z_z e m_z} \hat{\mu}_2^z I \frac{\partial V}{\partial \psi} \frac{T_z}{n_z} \frac{\partial}{\partial V} n_z \\ &\quad + \hat{\ell}_{11}^{ze} n_z \langle \textcolor{red}{Bu}_{e\parallel} \rangle + \hat{\ell}_{11}^{zi} n_z \langle \textcolor{red}{Bu}_{i\parallel} \rangle + \hat{\ell}_{11}^{zz} n_z \langle \textcolor{red}{Bu}_{z\parallel} \rangle + \hat{\ell}_{11}^{zb} \frac{n_z}{n_b} n_b \langle \textcolor{red}{Bu}_{b\parallel} \rangle \\ &\quad - \hat{\ell}_{12}^{ze} n_z \langle \textcolor{red}{B}\hat{q}_{e\parallel} \rangle - \hat{\ell}_{12}^{zi} n_z \langle \textcolor{red}{B}\hat{q}_{i\parallel} \rangle - \hat{\ell}_{12}^{zz} n_z \langle \textcolor{red}{B}\hat{q}_{z\parallel} \rangle \\ &\quad + Z_z e \frac{n_z}{m_z} I \langle R^{-2} \rangle \left(-\frac{\dot{\psi}_t}{q} + \dot{\psi} \right) \\ &\quad - \nu_L n_z \langle \textcolor{red}{Bu}_{z\parallel} \rangle - \nu_{0z} n_z \langle \textcolor{red}{Bu}_{z\parallel} \rangle - \nu_{\text{OL}} n_z \langle \textcolor{red}{Bu}_{z\parallel} \rangle \\ &\quad - \frac{\partial}{\partial V} \langle |\nabla \rho| \rangle v_{z\zeta} n_z \langle \textcolor{red}{Bu}_{z\parallel} \rangle + \langle |\nabla V| \rangle v_{z\zeta} n_z \left(\frac{\partial}{\partial V} \frac{\langle B^2 \rangle}{I} \right) \frac{I}{\langle B^2 \rangle} \langle \textcolor{red}{Bu}_{z\parallel} \rangle \\ &\quad + \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{z\zeta} n_z \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial V} \frac{I}{\langle B^2 \rangle} \langle \textcolor{red}{Bu}_{z\parallel} \rangle \\ &\quad - \langle |\nabla V|^2 \rangle \chi_{z\zeta} n_z \left(\frac{\partial}{\partial V} \frac{\langle B^2 \rangle}{I} \right) \frac{\partial}{\partial V} \frac{I}{\langle B^2 \rangle} \langle \textcolor{red}{Bu}_{z\parallel} \rangle - \frac{1}{m_z} \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial V} \langle \Pi_z^{\text{res}} \rangle \\ &\quad - \frac{\partial}{\partial V} \left[\langle |\nabla V| \rangle v_{z\zeta} n_z \frac{\langle R^2 \rangle}{\langle B^2 \rangle} \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{\partial \psi}{\partial V} (1 + 2\hat{q}^2) \frac{1}{I} \frac{\partial \psi}{\partial V} \textcolor{red}{BV}_{1z} \right. \\ &\quad \left. - \langle |\nabla V|^2 \rangle \chi_{z\zeta} n_z \frac{\partial}{\partial V} \frac{\langle R^2 \rangle}{\langle B^2 \rangle} \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{\partial \psi}{\partial V} (1 + 2\hat{q}^2) \frac{1}{I} \frac{\partial \psi}{\partial V} \textcolor{red}{BV}_{1z} \right] \end{aligned} \quad (3.24)$$

$$\begin{aligned} \frac{\partial}{\partial t} n_z \langle \textcolor{red}{Ru}_{z\zeta} \rangle &= -(u_z^V - u_g^V) \frac{\partial}{\partial V} n_z \langle \textcolor{red}{Ru}_{z\zeta} \rangle - \frac{\partial u_z^V}{\partial V} n_z \langle \textcolor{red}{Ru}_{z\zeta} \rangle - \frac{\partial}{\partial V} \langle |\nabla V| \rangle v_{z\zeta} n_z \langle \textcolor{red}{Ru}_{z\zeta} \rangle \\ &\quad + \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{z\zeta} \frac{\partial}{\partial V} n_z \langle \textcolor{red}{Ru}_{z\zeta} \rangle - \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{z\zeta} \langle \textcolor{red}{Ru}_{z\zeta} \rangle \frac{\partial}{\partial V} n_z - \frac{1}{m_z} \frac{\partial}{\partial V} \langle \Pi_z^{\text{res}} \rangle \\ &\quad + \hat{\ell}_{11}^{ze} \frac{n_z}{n_e} n_e \langle \textcolor{red}{Ru}_{e\zeta} \rangle + \hat{\ell}_{11}^{zi} \frac{n_z}{n_i} n_i \langle \textcolor{red}{Ru}_{i\zeta} \rangle + \hat{\ell}_{11}^{zz} n_z \langle \textcolor{red}{Ru}_{z\zeta} \rangle + \hat{\ell}_{11}^{zb} \frac{n_z}{n_b} n_b \langle \textcolor{red}{Ru}_{b\zeta} \rangle \\ &\quad - \hat{\ell}_{12}^{ze} n_z \frac{1}{e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_e - \hat{\ell}_{12}^{ze} n_z \frac{I}{\langle B^2 \rangle} \langle \textcolor{red}{B}\hat{q}_{e\parallel} \rangle \end{aligned}$$

$$\begin{aligned}
& + \hat{\ell}_{12}^{\text{zi}} n_z \frac{1}{Z_{\text{e}}} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\text{i}} - \hat{\ell}_{12}^{\text{zi}} n_z \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\text{i}\parallel} \rangle \\
& + \hat{\ell}_{12}^{\text{zz}} n_z \frac{1}{Z_{\text{e}}} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\text{z}} - \hat{\ell}_{12}^{\text{zz}} n_z \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\text{z}\parallel} \rangle \\
& + Z_{\text{z}} \frac{e}{m_{\text{z}}} n_z \dot{\psi} + Z_{\text{z}} \frac{e}{m_{\text{z}}} \frac{\partial \psi}{\partial V} n_{\text{z}} u_{\text{z}}^V \\
& + Z_{\text{z}} \frac{e \langle |\nabla V|^2 \rangle D_{\text{e}}}{m_{\text{z}} T_{\text{e}}} \frac{\partial \psi}{\partial V} \left[n_{\text{z}} \frac{\partial \psi}{\partial V} \frac{1}{I} B V_{1\text{e}} + n_{\text{z}} \frac{\partial}{\partial V} \Phi \right. \\
& \left. + \frac{C_{\text{T}} - 1}{e} \frac{n_{\text{z}}}{n_{\text{e}}} \left(\frac{\partial}{\partial V} n_{\text{e}} T_{\text{e}} - T_{\text{e}} \frac{\partial}{\partial V} n_{\text{e}} \right) + \frac{T_{\text{e}}}{e} \frac{\partial \psi}{\partial V} C_{\text{P}} n_{\text{z}} \right] \\
& - \nu_{\text{L}} n_{\text{z}} \langle R u_{\text{z}\zeta} \rangle - \nu_{0\text{z}} n_{\text{z}} \langle R u_{\text{z}\zeta} \rangle - \nu_{\text{OL}} n_{\text{z}} \langle R u_{\text{z}\zeta} \rangle
\end{aligned} \tag{3.25}$$

$$\begin{aligned}
\frac{3}{2} \frac{\partial}{\partial t} n_{\text{z}} T_{\text{z}} & = -\frac{5}{2} (u_{\text{z}}^V - u_{\text{g}}^V) \frac{\partial}{\partial V} n_{\text{z}} T_{\text{z}} - \frac{5}{2} \frac{\partial u_{\text{z}}^V}{\partial V} n_{\text{z}} T_{\text{z}} - \frac{\partial}{\partial V} V_{\text{z}}^{\text{hp}} \langle |\nabla V| \rangle n_{\text{z}} T_{\text{z}} \\
& + \frac{\partial}{\partial V} \chi_{\text{z}} \langle |\nabla V|^2 \rangle \frac{\partial}{\partial V} n_{\text{z}} T_{\text{z}} - \frac{\partial}{\partial V} \chi_{\text{z}} \langle |\nabla V|^2 \rangle T_{\text{z}} \frac{\partial}{\partial V} n_{\text{z}} \\
& + (u_{\text{z}}^V - u_{\text{g}}^V) \frac{\partial}{\partial V} n_{\text{z}} T_{\text{z}} + \hat{u}_{\text{z}\theta} \langle \mathbf{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_{\text{z}} \rangle - \frac{3}{2} \nu_{\text{Tez}} \left(\frac{n_{\text{e}}}{n_{\text{z}}} n_{\text{z}} T_{\text{z}} - n_{\text{e}} T_{\text{e}} \right) - \frac{3}{2} \nu_{\text{Tiz}} \left(\frac{n_{\text{i}}}{n_{\text{z}}} n_{\text{z}} T_{\text{z}} - n_{\text{i}} T_{\text{i}} \right) \\
& - \nu_{\text{L}} (n_{\text{z}} T_{\text{z}} - n_{\text{zsup}} T_{\text{z}}) - \frac{3}{2} \nu_{\text{LTz}} (n_{\text{z}} T_{\text{z}} - n_{\text{z}} T_{\text{zsup}}) + P_{\alpha\text{z}} + P_{\text{RFz}}
\end{aligned} \tag{3.26}$$

$$\begin{aligned}
\frac{5}{2} n_{\text{z}} \frac{\partial}{\partial t} \langle B \hat{q}_{\text{z}\parallel} \rangle & = -\frac{1}{m_{\text{z}}} \hat{\mu}_2^{\text{z}} \langle B u_{\text{z}\parallel} \rangle - \frac{1}{m_{\text{z}}} \hat{\mu}_3^{\text{z}} \langle B \hat{q}_{\text{z}\parallel} \rangle + \frac{1}{m_{\text{z}}} \hat{\mu}_2^{\text{z}} B V_{1\text{z}} \\
& - \frac{1}{Z_{\text{z}} e m_{\text{z}}} \hat{\mu}_3^{\text{z}} I \frac{\partial V}{\partial \psi} \frac{1}{n_{\text{z}}} \frac{\partial}{\partial V} n_{\text{z}} T_{\text{z}} + \frac{1}{Z_{\text{z}} e m_{\text{z}}} \hat{\mu}_3^{\text{z}} I \frac{\partial V}{\partial \psi} \frac{T_{\text{z}}}{n_{\text{z}}} \frac{\partial}{\partial V} n_{\text{z}} \\
& - \hat{\ell}_{21}^{\text{ze}} n_{\text{z}} \langle B u_{\text{e}\parallel} \rangle - \hat{\ell}_{21}^{\text{zi}} n_{\text{z}} \langle B u_{\text{i}\parallel} \rangle - \hat{\ell}_{21}^{\text{zz}} n_{\text{z}} \langle B u_{\text{z}\parallel} \rangle - \hat{\ell}_{21}^{\text{zb}} \frac{n_{\text{z}}}{n_{\text{b}}} n_{\text{b}} \langle B u_{\text{b}\parallel} \rangle \\
& + \hat{\ell}_{22}^{\text{ze}} n_{\text{z}} \langle B \hat{q}_{\text{e}\parallel} \rangle + \hat{\ell}_{22}^{\text{zi}} n_{\text{z}} \langle B \hat{q}_{\text{i}\parallel} \rangle + \hat{\ell}_{22}^{\text{zz}} n_{\text{z}} \langle B \hat{q}_{\text{z}\parallel} \rangle \\
& - \nu_{\text{L}} n_{\text{z}} \langle B \hat{q}_{\text{z}\parallel} \rangle - \nu_{0\text{z}} n_{\text{z}} \langle B \hat{q}_{\text{z}\parallel} \rangle
\end{aligned} \tag{3.27}$$

$$0 = -n_{\text{z}} \left\langle \frac{u_{\text{z}\zeta}}{R} \right\rangle + \frac{1}{\langle R^2 \rangle} n_{\text{z}} \langle R u_{\text{z}\zeta} \rangle - \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{\langle R^2 \rangle} n_{\text{z}} \langle R u_{\text{z}\zeta} \rangle + \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{I} n_{\text{z}} \langle B u_{\text{z}\parallel} \rangle \tag{3.28}$$

$$0 = -n_{\text{z}} B V_{1\text{z}} - \frac{1}{Z_{\text{z}} e} \frac{\partial V}{\partial \psi} I \frac{\partial}{\partial V} n_{\text{z}} T_{\text{z}} - n_{\text{z}} \frac{\partial V}{\partial \psi} I \frac{\partial}{\partial V} \Phi \tag{3.29}$$

$$\begin{aligned}\frac{\partial}{\partial t}n_b &= -\frac{\partial}{\partial V}n_b u_b^V + u_g^V \frac{\partial}{\partial V}n_b + \frac{P_b}{E_b} - g \frac{P_{b\perp}}{E_b} - \nu_b n_b - \nu_{bL} n_b - \nu_{\text{eff}} g(\sqrt{\delta} n_b - n_b^{\text{rp}}) \\ &+ \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle f_t D_b \frac{\partial}{\partial V} n_b\end{aligned}\quad (3.30)$$

$$0 = Z_b \frac{e}{m_b} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{1}{I} n_b B V_{1b} + Z_b \frac{e}{m_b} I n_b \langle B u_{b\parallel} \rangle - Z_b \frac{e}{m_b} \langle B^2 \rangle n_b \langle R u_{b\zeta} \rangle \quad (3.31)$$

$$\begin{aligned}\frac{\partial}{\partial t} n_b \langle B u_{b\parallel} \rangle &= -\frac{1}{m_b n_b} \hat{\mu}_1^b n_b \langle B u_{b\parallel} \rangle + \frac{1}{m_b n_b} \hat{\mu}_1^b n_b B V_{1b} \\ &+ \hat{\ell}_{11}^{\text{bb}} \frac{m_e n_e}{m_b n_b} n_b \langle B u_{b\parallel} \rangle + \sum_{a \neq b} \hat{\ell}_{11}^{\text{ba}} \frac{m_e n_e}{m_b} \langle B u_{a\parallel} \rangle \\ &+ Z_b \frac{e}{m_b} n_b I \langle R^{-2} \rangle \left(-\frac{\dot{\psi}_t}{q} + \dot{\psi} \right) - \nu_{bL} n_b \langle B u_{b\parallel} \rangle - \nu_b n_b \langle B u_{b\parallel} \rangle - \nu_{0b} n_b \langle B u_{b\parallel} \rangle \\ &- \nu_{\text{CX}} n_b \langle B u_{b\parallel} \rangle + \frac{1}{m_b} \langle B S_{\text{m}\parallel b} \rangle - \nu_{\text{eff}} g \sqrt{\delta} n_b \langle B u_{b\parallel} \rangle + \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \langle B u_{b\parallel} \rangle f_t D_b \frac{\partial}{\partial V} n_b\end{aligned}\quad (3.32)$$

$$\begin{aligned}\frac{\partial}{\partial t} n_b \langle R u_{b\zeta} \rangle &= u_g^V \frac{\partial}{\partial V} n_b \langle R u_{b\zeta} \rangle - \frac{\partial}{\partial V} \langle R u_{b\zeta} \rangle n_b u_b^V \\ &+ \frac{1}{m_b} \ell_{11}^{\text{be}} \frac{1}{n_e} n_e \langle R u_{e\zeta} \rangle + \frac{1}{m_b} \ell_{11}^{\text{bi}} \frac{1}{n_i} n_i \langle R u_{i\zeta} \rangle + \frac{1}{m_b} \ell_{11}^{\text{bz}} \frac{1}{n_z} n_z \langle R u_{z\zeta} \rangle + \frac{1}{m_b} \ell_{11}^{\text{bb}} \frac{1}{n_b} n_b \langle R u_{b\zeta} \rangle \\ &+ Z_b \frac{e}{m_b} n_b \dot{\psi} + Z_b \frac{e}{m_b} \frac{\partial \psi}{\partial V} n_b u_b^V \\ &- \nu_{bL} n_b \langle R u_{b\zeta} \rangle - \nu_b n_b \langle B u_{b\zeta} \rangle - \nu_{0i} n_b \langle R u_{b\zeta} \rangle - \nu_{\text{CX}} n_b \langle R u_{b\zeta} \rangle + \frac{1}{m_b} \frac{I}{\langle B^2 \rangle} \langle B S_{\text{m}\parallel b} \rangle\end{aligned}\quad (3.33)$$

$$0 = -n_b \left\langle \frac{u_{b\zeta}}{R} \right\rangle + \frac{1}{\langle R^2 \rangle} n_b \langle R u_{b\zeta} \rangle - \frac{2\hat{q}^2}{1+2\hat{q}^2} \frac{1}{\langle R^2 \rangle} n_b \langle R u_{b\zeta} \rangle + \frac{2\hat{q}^2}{1+2\hat{q}^2} \frac{1}{I} n_b \langle B u_{b\parallel} \rangle \quad (3.34)$$

$$0 = -n_b B V_{1b} - \frac{1}{Z_b e} \frac{\partial V}{\partial \psi} I n_b \frac{\partial}{\partial V} T_b - \frac{1}{Z_b e} \frac{\partial V}{\partial \psi} I T_b \frac{\partial}{\partial V} n_b - n_b \frac{\partial V}{\partial \psi} I \frac{\partial}{\partial V} \Phi \quad (3.35)$$

$$\begin{aligned}\frac{\partial}{\partial t} n_{01} &= -\frac{\partial u_g^V}{\partial V} n_{01} + \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle D_{01} \frac{\partial}{\partial V} n_{01} - \nu_{\text{ionize}} \frac{n_e}{n_{01} + n_{02} + n_{03}} n_{01} \\ &- \nu_{\text{CX}} \frac{n_i}{n_{01} + n_{02}} n_{01} + \gamma_0 \nu_L (n_i - n_{\text{isup}}) + \frac{\partial}{\partial V} V' \Gamma^{\text{puff}} \Big|_b\end{aligned}\quad (3.36)$$

$$\frac{\partial}{\partial t} n_{02} = -\frac{\partial u_g^V}{\partial V} n_{02} + \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle D_{02} \frac{\partial}{\partial V} n_{02} - \nu_{\text{ionize}} \frac{n_e}{n_{01} + n_{02} + n_{03}} n_{02} + \nu_{\text{CX}} \frac{n_i}{n_{01} + n_{02}} n_{01} \quad (3.37)$$

$$\frac{\partial}{\partial t} n_{03} = -\frac{\partial u_g^V}{\partial V} n_{03} + \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle D_{03} \frac{\partial}{\partial V} n_{03} - \nu_{\text{ionize}} \frac{n_e}{n_{01} + n_{02} + n_{03}} n_{03} + f_{\text{CX}} \frac{P_b}{E_b} \quad (3.38)$$

$$\begin{aligned}
\frac{\partial}{\partial t} \bar{n}_z &= -\frac{\partial u_g^V}{\partial V} \bar{n}_z + \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle D_{0z} \frac{\partial}{\partial V} \bar{n}_z - n_e \bar{n}_z f_5 S_5 + n_e n_z A_6 \\
&\quad + \gamma_{0z} \nu_L (n_z - n_{z\text{sup}}) + \gamma_{i,z} \nu_L (n_i - n_{i\text{sup}}) + \left. \frac{\partial}{\partial V} V' \Gamma_z^{\text{puff}} \right|_{\text{b}}
\end{aligned} \tag{3.39}$$

$$\begin{aligned}
\frac{\partial}{\partial t} n_{\text{b}}^{\text{rp}} &= -\frac{\partial u_g^V}{\partial V} n_{\text{b}}^{\text{rp}} \frac{P_{\text{b}\perp}}{E_{\text{b}}} + \nu_{\text{eff}} (\sqrt{\delta} n_{\text{b}} - n_{\text{b}}^{\text{rp}}) - \nu_{\text{b}} n_{\text{b}}^{\text{rp}} \\
&\quad - \frac{\partial}{\partial V} \langle |\nabla V| \rangle u_{\text{b}}^{\text{rp}} n_{\text{b}}^{\text{rp}} + \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle D_{\text{b}}^{\text{rp}} \frac{\partial}{\partial V} n_{\text{b}}^{\text{rp}}
\end{aligned} \tag{3.40}$$

4 Boundary Conditions

num.	name	variable	highest order derivative	center	edge
1	LQm1	Φ	2	N	0
2	LQm2	$\dot{\psi}_t$	2	0	$\langle \nabla V ^2 / R^2 \rangle _{V_b} I_{\text{vac}} / (4\pi^2)$
3	LQm3	$\dot{\psi}$	2	N	$2\pi\mu_0 I_p$
4	LQm4	ψ	0	$\times$	$\times$
5	LQm5	ψ_t / μ_0	0	$\times$	$\times$
6	LQe1	n_e	2	N	N
7	LQe2	$n_e u_e^V$	1	0	$\times$
8	LQe3	$\langle Bu_{e\parallel} \rangle$	2(0)	N($\times$)	N($\times$)
9	LQe4	$n_e \langle Ru_{e\zeta} \rangle$	2	N	N
10	LQe5	$n_e T_e$	2	N	N
11	LQe6	$\langle B\hat{q}_{e\parallel} \rangle$	0	$\times$	$\times$
12	LQe7	$n_e \langle u_{e\zeta} / R \rangle$	0	$\times$	$\times$
13	LQe8	BV_{1e}	0	$\times$	$\times$
14	LQi1	n_i	2	N	N
15	LQi2	$n_i u_i^V$	1	0	$\times$
16	LQi3	$\langle Bu_{i\parallel} \rangle$	2(0)	N($\times$)	N($\times$)
17	LQi4	$n_i \langle Ru_{i\zeta} \rangle$	2	N	N
18	LQi5	$n_i T_i$	2	N	N
19	LQi6	$\langle B\hat{q}_{i\parallel} \rangle$	0	$\times$	$\times$
20	LQi7	$n_i \langle u_{i\zeta} / R \rangle$	0	$\times$	$\times$
21	LQi8	BV_{1i}	0	$\times$	$\times$
22	LQz1	n_z	2	N	N
23	LQz2	$n_z u_z^V$	1	0	$\times$
24	LQz3	$\langle Bu_{z\parallel} \rangle$	2(0)	N($\times$)	N($\times$)
25	LQz4	$n_z \langle Ru_{z\zeta} \rangle$	2	N	N
26	LQz5	$n_z T_z$	2	N	N
27	LQz6	$\langle B\hat{q}_{z\parallel} \rangle$	0	$\times$	$\times$
28	LQz7	$n_z \langle u_{z\zeta} / R \rangle$	0	$\times$	$\times$
29	LQz8	BV_{1z}	0	$\times$	$\times$
30	LQb1	n_b	0(2)	$\times$ (N)	$\times$ (N)
31	LQb2	$n_b u_b^V$	1	0	$\times$
32	LQb3	$n_b \langle Bu_{b\parallel} \rangle$	0	$\times$	$\times$
33	LQb4	$n_b \langle Ru_{b\zeta} \rangle$	0	$\times$	$\times$
34	LQb7	$n_b \langle u_{b\zeta} / R \rangle$	0	$\times$	$\times$
35	LQb8	BV_{1b}	0	$\times$	$\times$
36	LQn1	n_{01}	2	N	$\langle \nabla V \rangle S_{\text{gas}} _b$
37	LQn2	n_{02}	2	N	0
38	LQn3	n_{03}	2	N	0
39	LQnz	$\bar{n}_z$	2	N	$\langle \nabla V \rangle S_{\text{gas},z} _b$
40	LQr1	n_b^{rp}	0(2)	$\times$ (N)	$\times$ (N)

0: Dirichlet condition ($u = 0$), N: Neumann condition ($u' = 0$)

value: Neumann condition ($u' = \text{value}$), $\times$: no condition

NOTE: The dependent variable of LQm4 turns to be ψ/μ_0 when NB is applied. Subsequently, the boundary condition of LQm3 is also changed to $2\pi I_p$.

NOTE: In the table the column labeled by the highest order derivative means the order of the highest order derivative of a dependent variable, not an equation, in the corresponding row. Please note that the term not integrated by parts is never counted. For example, the first row shows the highest order of derivatives is equal to 2 for Φ because the LQm1 equation has the diffusion term having the second derivative of Φ with respect to V . In addition, the LQe2 and LQi2 equations also have the first derivative of Φ , which is not a dependent variable of these equations. As a whole, the highest order derivative of Φ in this equation system is 2 that comes from the LQm1 equation. When we look at n_e , we find that the highest order derivative of n_e is 2 shown in the LQe5 equation, even though no derivative appears in the LQe1 equation. The Neumann conditions for n_e come from the LQe5 equation. When it comes to $n_e u_e^V$, the highest derivative is found as order 1 in the LQe1 equation, but this term is not integrated by parts. It means that any boundary condition is not naturally imposed on $n_e u_e^V$. The Dirichlet boundary condition should be imposed at axis.